Operation Manual

MCR-F110

Fixed Code Reader

July 7, 2023

MARS TOHKEN SOLUTION CO. LTD.

3rd Edition

Record of Changes

Edition	Date	Description of Change
1	December 20, 2019	New release
2	March 22, 2022	Add "Save NG (No Read) Image" Add Erase the setting of SETUP Add SETUP number addition
3	July 7, 2023	Compatible with firmware version V1.3a 7.11 PLC Link Add "Panasonic FP7 series" 7.17 General Operation Add "Reboot the reader."

Introduction

Thank You For Your Purchase.

This manual explains the specifications, setup, Installation, features, operation, and system configuration, of the MCR-F110 code reader.

In order to use the product properly, read this manual carefully.

If you encounter any issue, please describe the detailed step to reproduce it for our investigation.

The contents of this manual may change without notice. Check our website for the most current version.

Safety notices



DO NOT disassemble. Disassembly will void the warranty and could cause damage or personal injury.



This product is intended for use in general electronics equipment (electronic computer, OA equipment, communications equipment, test and measurement equipment, machine tools, industrial robots, AV equipment, home appliances etc.).
Appropriate measures should be taken at use of unit/system related to safety and operation of transportation equipment (aircraft, train, automobile), traffic signal, gas leak detector, and safety devices.



MARS TOHKEN SOLUTION products are not designed or intended for the use as a component in life support appliances or as surgical implants nor in any other application where failure of the product could cause personal injury or death. The use of MARS TOHKEN SOLUTION products shall indemnify and hold harmless to MARS TOHKEN SOLUTION's and its shareholders against any claims of injury or death associated with unintended authorized use.

Handle with care



- DATA BACKUP

This product has a memory backup function. This backup data restoration cannot be guaranteed if repair, reconstruction, or upgrades are performed on this product.

- DO NOT use this product at temperature or humidity ranges beyond that documented in the product specifications, or in direct sunlight.
- DO NOT expose this product to water, moisture, oil, etc.
- This product may be damaged in environments containing corrosive gas.
- DO NOT use any chemicals when cleaning.

If there are any dust particles on the image sensor, please clean the sensor as follows:

- Wipe lightly with cloth or swab (may be damp with alcohol).
- Wipe off any residual alcohol using a dry cloth.
- This is a high-precision optical device, avoid exposing this product to excessive force such as that sustained by a drop.



WARNING

Do not install electrical wiring or plugin/unplug of cable when this product is powered on. These may result in an electrical damage to this product.

To minimize the risk of data loss, make sure to backup the hard drive before installing the software. Our company shall not be responsible for any troubles such as data loss or damages.

Locations for operation and storage

Avoid the following locations that could cause an accident or damage to the product.

- Exposed to ambient temperature outside the rating
- Exposed to relative humidity outside the rating
- Exposed to rapid temperature fluctuations (causing condensation)
- Exposed to direct sunlight or near heaters
- Exposed to direct vibration or shock
- In the presence of dust, salt, or iron particles
- In environments where static electricity can build into significant charges
- In the presence of flammable or explosive gases
- In the presence of corrosive gas

Long-term storage

Store away from direct sunlight and direct vibration or shock for long-term storage. Do not store this product at temperature or humidity ranges beyond that documented in the product specifications.

Precautions on Regulation and Standards

CE Marking

MARS TOHKEN SOLUTION CO.LTD. has confirmed that this product complies with the essential requirements of the applicable EMC Directive and EU RoHS Directive, based on the following specifications. Be sure to consider the following specifications when using this product in a Member State of European Union.

●EMC Directive (2014/30/EU)

These specifications do not give any guarantee that the end-product with this product incorporated complies with the essential requirements of EMC Directive. The manufacturer of the end-product is solely responsible for the compliance on the end-product itself according to EMC Directive.

●EU RoHS Directive (2011/65/EU)

FCC Regulations

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.



This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

China RoHS

产品中有害物质的名称及含量

部件名称	有毒有害物质或元素					
	铅 (Pb)	汞 (Hg)	镉 (Cd)	六价铬 (Cr(VI))	多溴联苯 (PBB)	多溴二苯醚 (PBDE)
实装電路板	×	O	O	O	O	O
外装部件(*)	×	O	O	O	O	O
电缆	×	O	O	O	O	O
包装/标签	O	O	O	O	O	O

本表格依据 SJ/T11364 的规定编制。

○： 表示该有害物质在该部件所有均质材料中的含量均在 GB/T26572 规定的限量要求以下。

×： 表示该有害物质至少在该部件的某一均质材料中的含量超出 GB/T26572 规定的限量要求。

(*)外装部件包括外廓壳体、光学系零件、配线用螺丝、端子、安装支架等零件。

●产品污染控制标识

该标识适用于在中华人民共和国境内销售的电子信息产品. 本产品的使用年限是组装到整机中，通常情况下能使用的年限.

图形中间的数字表示电子信息产品的环保使用期限.

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电话:021-62414845

●识别生产日期的方法

请通过产品外包装箱上的序列号识别生产日期.

如) 在表示生产日期为2019年1月



Taiwan RoHS

設備名稱：固定式條碼閱讀機 ， 型號（型式）：MCR-F110 系列
Equipment name: Fixed Code Reader , Type designation (Type): MCR-F110 series

單元 Unit	限用物質及其化學符號 Restricted substances and its chemical symbols					
	鉛 Lead (Pb)	汞 Mercury (Hg)	鎘 Cadmium (Cd)	六價鉻 Hexavalent chromium (Cr^{+6})	多溴聯苯 Polybrominated Biphenyls (PBB)	多溴二苯醚 Polybrominated diphenyl ethers (PBDE)
電路板	–	○	○	○	○	○
外殼	-	○	○	○	○	○
电缆	-	○	○	○	○	○
包装/标签	○	○	○	○	○	○

備考1.〝超出0.1wt %〞及〝超出0.01wt %〞係指限用物質之百分比含量超出百分比含量基準值。

Note 1：“Exceeding 0.1 wt %” and “exceeding 0.01 wt %” indicate that the percentage content of the restricted substance exceeds the reference percentage value of presence condition.

備考2.〝○〞係指該項限用物質之百分比含量未超出百分比含量基準值。

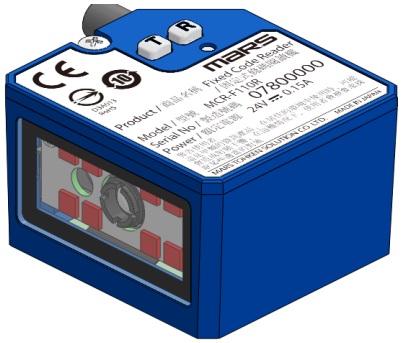
Note 2：“O” indicates that the percentage content of the restricted substance does not exceed the percentage of reference value of presence.

備考3.〝 - 〞係指該項限用物質為排除項目。

Note 3：The “–” indicates that the restricted substance corresponds to the exemption.

BSMI EMC Notice

警告使用者：
這是甲類的資訊產品，在居住的環境中使用時，可能會造成射頻干擾，在這種情況下，使用者會被要求採取某些適當的對策。



Download information

The user manuals, software, and device drivers are available from:

https://www.mars-tohken.co.jp/en/DL/

1. Operation Manual: This document.
2. MTS USB Serial COM Port: Driver software for USB Virtual COM.
3. MCRConfig: Communication & configuration software.
4. PLC Link Instruction Manual for Image Reader: Describes how to use PLC-Link.
5. 2D code menu : Configuration codes of MCR-F110.



WARNING

The following conditions are recommended when printing the operation manual.
[black-and-white printing on A5 size plain papers, more than 600DPI resolution]
Resolutions lower than 600 DPI may cause reading errors of the reader's setting symbols in Section 2.2.

Unpacking

Check that the box includes the items listed below. Please contact our sales representative if you find any missed or damaged item.

- Contents

Item	Quantity
MCR-F110 code reader	1
Quick start guide	1
China RoHS ・Taiwan RoHS Details	1

Optional item

- Optional item for MCR-F110R

Item	Model name
Power Supply Unit	MPSU-3200-24V
MPSU-3200 relay cable	AA000343

Table of Contents

Record of Changes **1**

1 Overview **12**

1.1 Supported symbologies 13
1.2 Product lineup 14
1.3 Product description 15

2 Preparation **19**

2.1 RS-232C type 19
2.2 USB type 19
2.3 Reading flow (Note: in case of default setting) 21
2.4 Configuration 21
2.5 Image data transfer 22

3 Operation Mode **23**

3.1 Single Reading Mode 24
3.2 Timeout Mode 26
3.3 External Trigger Mode 30
3.4 Continuous Reading Mode 31
3.5 Test Mode 31
3.6 Cycle Buffer Function 33
3.7 Concept of Cycle Buffer 34

4 Configuration for symbol reading **35**

4.1 Automatic parameter tuning (Auto-Tuning) 35
4.2 Table Mode [AGC=T] 39
4.3 Setting parameters for Table Mode 41
4.4 Auto Gain Control Mode [AGC=E] 45
4.5 Reading parameters for Auto Gain Control Mode 45

5 Auto-sense Mode **47**

5.1 How to use 47
5.2 Timing chart 48
5.3 Setting parameters for Auto-sense Mode 48

6 Advanced Features **50**

6.1 Preset Mode 50

6.2 Output Additional Information 53
6.3 Save Image 58
6.4 Save NG (No Read) Image 60
6.5 PLC Link [Only for RS-232C type] 62
6.6 Simultaneous reading of multiple labels 62
6.7 Image Rotation 62
6.8 Register of SYNC ON/OFF command 63
6.9 Save SETUP / Load SETUP 64

7 Serial Commands (RS-232C, USB) 71

7.1 Communication settings 72
7.2 Symbologies settings 73
7.3 Symbol Reading (Operation, Adjustment and Diagnostic mode) 79
7.4 Auto-Tuning 81
7.5 Camera Control (1) 81
7.6 Camera Control (2) (for Table Mode) 81
7.7 Camera Control (3) (for Table Mode) 82
7.8 Camera Control (3) (for Auto Gain Control Mode) 83
7.9 Image Preprocessing 84
7.10 Preset Mode 84
7.11 PLC Link 85
7.12 Auto sense 86
7.13 Image Output 86
7.14 Save NG (No Read) Image 87
7.15 Configuration Reference 87
7.16 Setup 87
7.17 General Operation 88
7.18 Table of Character Code 88

8 Specifications 89

8.1 Specifications 89
8.2 Reading Specifications 91
8.3 Dimensions 92
8.4 Interface 94

9 Peripherals 97

9.1 Peripheral for MCR-F110R 97

10 Troubleshooting 99

10.1 MCR-F110 does not start up or cannot communicate with a PC. 99
10.2 Symbol cannot be decoded 100
10.3 Others 101

1 Overview

(1) MCR-F110 is a fixed mount (stationary type) code reader capable of 1-dimensional (1D) / 2-dimensional (2D) codes reading. This unit incorporates the most innovative digital camera technologies, image processing, and software techniques. This manual may also refer to 1D / 2D codes as ”symbols”.

(2) MCR-F110 is a compact reader that can be mounted in a narrow space. IP65 enclosure allow use in harsh environments.

(3) MCR-F110’s following features help to read challenging direct marking by laser maker or dot peen.

- Image Preprocessing:
  From raw image, MCR-F110 can modify it by its image preprocess technique for better reading ratio or reading speed.
- Table Mode:
  Maximum 16 sets of configuration can be used seamlessly. This mode attempts to read by changing the set of configuration one by one until it get successful decoding or within the limit of time.

(4) Cycle Buffer Function (MAXIMG) performs both the decoding and capturing images in parallel. This function eliminates the detailed timing tuning for moving symbols.

1.1 Supported symbologies

1D barcode

Code39
Code128 / GS1-128 (EAN128)
Codabar (NW7)
ITF (Interleaved 2 of 5)
JAN / EAN / UPC
GS1 DataBar (RSS)
Code93

2D code

Data Matrix (ECC200)
QR Code / Micro QR Code
PDF417 / MicroPDF
Aztec Code
Composite Code

1.2 Product lineup

MCR-F110 consists of 2 models in interface classification.

Model name	Interface	Cable length
MCR-F110R	RS-232C	2m
MCR-F110U	USB (Full speed)	2m

(1) RS-232C type

Regarding RS-232C type, RS-232C, digital I/O (Input x1, Output x2) can be used.
The power supply is DC 24V.

(2) USB type

Regarding USB type, two types of operating modes can be used.
Please refer to Section 2.2 for instructions for switching the operating modes.

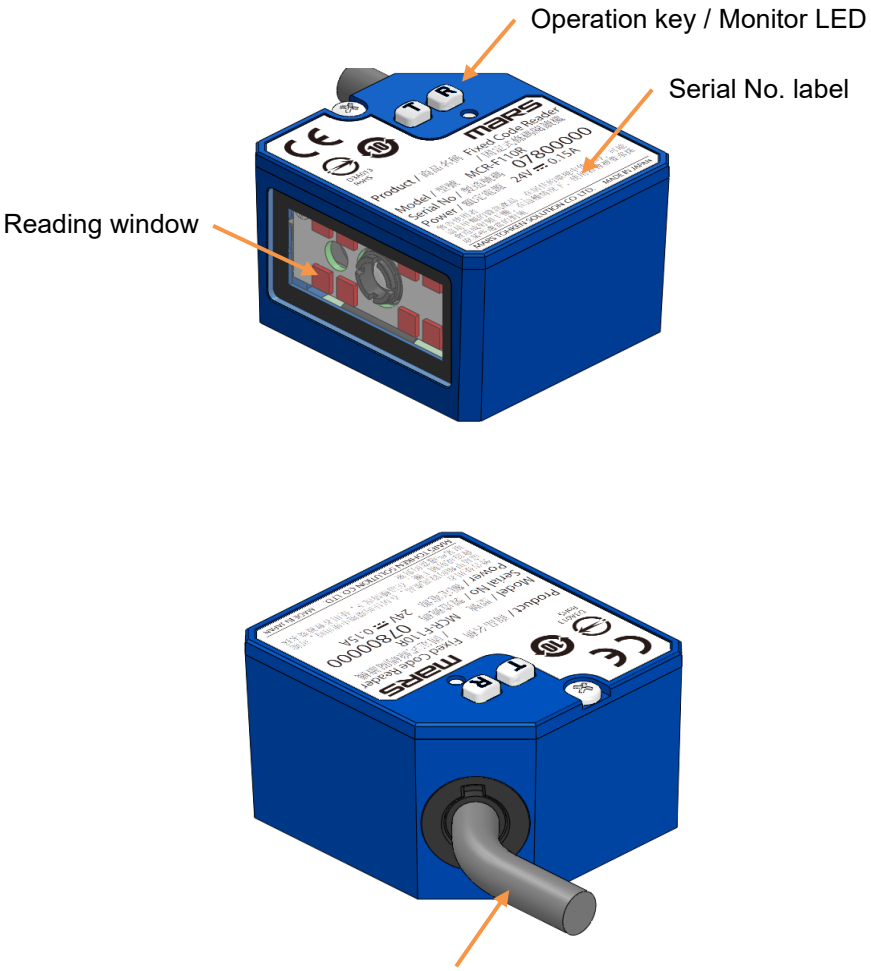
Virtual COM mode:

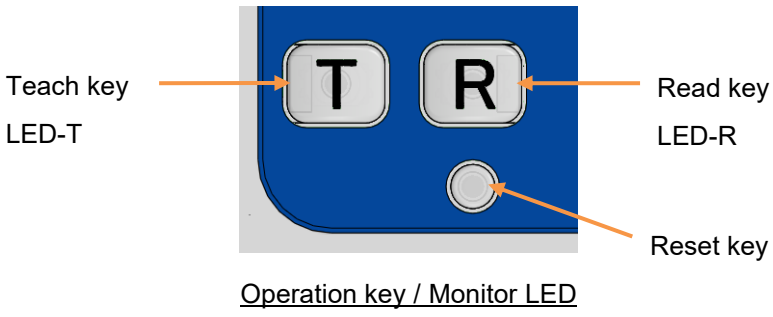
USB serial communication is available by installing the USB driver “MTS USB Serial COM Port” on the host PC. Depending on the condition of peripherals of the PC, some special handlings can be required.

HID (Human Interface Device) mode:

The reader can be used as a keyboard emulation device. The reading data is transmitted as if it was an input from the keyboard.

1.3 Product description





Operation key / Monitor LED

1.3.1 **Operation key**

- Teach key
  For Auto-Tuning function. In addition, switching Virtual COM mode and HID mode alternately by turning on the power while pressing the key (only for USB type).

- Read key
  For a test / verification and Auto-Tuning function, and also trigger input same like the external synchronizing signal.

- Reset key
  For restarting the reader.

1.3.2 **Monitor LED (LED-T, LED-R)**

The status is displayed by the LED color and lighting pattern of the operation key.

Status	Monitor LED	
Power on	RS-232C： USB(COM)： USB(HID)：	Turns red⇒green⇒off Turns red⇒green⇒off Turns red⇒orange⇒off
Reading symbols	OK： NG：	Turns green Turns red
Auto-Tuning	In progress： Success： Failure： Save - 3 tables： - 1 table：	Blinks orange Turns green Blinks red 7 times Blinks green 4 times Blinks green 3 times
PLC link	Blinks red	
Camera module error	Blinks red at short intervals	
Connection error / Communication error	Blinks red 7 times at short intervals	
Save settings	Blinks green 3 times	

1.3.3 **Interface cable**

RS-232C type

- Power
  Supply DC24V to the reader.

- RS-232C interface: RxD, TxD, CTS, RTS
  Used for transmitting the decoded data and status and receiving commands from host device.

- 2 Digital output: OUT1, OUT2
  These are photocoupler isolated digital signal outputs.

  OUT1 and OUT2 outputs statuses of the reading operation and the reading results. Allocation and timing setting for each output can be configured by serial commands (refer to 7.17 ”General Operation”).

Default settings of status assignments

Pin	Default
OUT1	GO signal
OUT2	NG signal

Status signals

Signal	Description
GO signal	GO signal is associated with Read OK, which is ON during the specified time when the symbol is decoded successfully. And also this signal is OFF when starting the next reading.
NG signal	NG signal is associated with Read NG, which is ON during the specified time when the symbol is not decoded successfully. And also this signal is OFF when starting the next reading.
Ready signal	Ready signal is ON when the code reader is ready for reading.
Busy1 signal	Busy1 signal is ON during a reading (except switch chattering delay), and OFF after decoding.
Busy2 signal	Busy2 signal is ON during a reading (except switch chattering delay), and OFF after transmitting decoded data.



When changing these parameters, save the settings to the internal flash memory (using the WSETS command) and restart the code reader.
When using Busy2 signal, the settings RSCS=1 (Use RS/CS control).

- Digital input: SYNC
  This signal is photocoupler isolated digital input used to trigger the reading action.

USB type

Select the device type (Virtual COM mode / HID mode).

2 Preparation

2.1 RS-232C type

(1) Connect with host device. (Refer to 9.1.2 “Example of device configuration”)

(2) Supply power to the reader.

Default settings of RS-232C communication

Baud rate	: 9600bps
Data length	: 8bit
Parity	: None
Stop bit	: 1bit
Flow control	: None

2.2 USB type

The USB type can be switched between Virtual COM mode and HID mode. The mode switching method is as follows. Default setting is Virtual COM mode.

- Turn on the power while pressing the Teach key. When the monitor LED changes from red to green or orange, release the key.
- Read the following symbols in order of (1) ”Start the mode switching” ⇒ (2) ”the mode you want to switch to” ⇒ (3) ”Save the new mode”, then reboot the reader.



Start the mode switching



Virtual COM mode



HID mode



Save the new mode
(Save to flash ROM)

2.2.1 **USB Virtual COM mode**

(1) Check if USB port of the Windows PC is available.

(2) Download the USB driver “MTS USB Serial COM Port” from our website.
https://www.mars-tohken.co.jp/en/DL/

(3) Refer to the instructions included in the downloaded file and install the driver. Normally, those operations will be done only one time.

(4) Plug the USB connector into an available USB port while the PC is running. The monitor LED lights (red ⇒ green ⇒ off) during start-up.

(5) Check the PC can receive the symbol data successfully by using communication software such as Hyper Terminal or “MCRConfig” which is communication and configuration software for the reader.

2.2.2 **HID mode**

(1) Check if USB port of the Windows PC is available.

(2) Plug the USB connector into an available USB port while the PC is running. The monitor LED lights (red ⇒ orange ⇒ off) during start-up.

(3) The PC will recognize the reader as human interface device.

(4) The PC will processes the symbol data in the same way as the data entered with the keyboard.

2.3 Reading flow (Note: in case of default setting)

(1) Input the trigger for reading start: 3 ways to do

- The serial command “g”
- The Read key of operation keys
- The digital input “SYNC input” (Only for RS-232C type)

When trigger is input, the monitor LED turns green and the reading operation starts.

(2) Return the result: 2 ways

a. Read OK

The decoded data is transmitted through the interface.

RS-232C type

- Digital output “GO” is ON and the monitor LED turns green.

USB type

- The monitor LED turns green.

b. Read NG

The error code is transmitted through the interface.

RS-232C type

- Digital output “NG” is ON and the monitor LED turns red.

USB type

- The monitor LED turns red.

2.4 Configuration

Configure settings of the reader by serial commands. However, the serial commands can not be used in USB HID mode. When sending the serial commands, use in USB Virtual COM mode.

For some settings, you can use the 2D code menu for settings.

2.5 Image data transfer

The image data that the reader captured can be transferred to the host PC using the configuration tool “MCRConfig”. By changing the setting, it is possible to transfer when the reader is in standby state or when reading process is complete.

Estimated transmission time of image data

Conditions: Bit map format (file name *.bmp, 752 x 480 pixels)

RS-232C type Baud rate 115.2kbps: Approx. 30 seconds

USB type USB2.0 (Full Speed): Approx. 1 second

Command	Description
MODE=0	Not sending data. (*)
MODE=1	Test mode: DO NOT USE
MODE=2	
MODE=3	Transfers image data every time (only for USB type)
MODE=4	Transfers image data when the reading is “NG”. (only for USB type)
MODE=5	Transfers image data when the reading is “OK”. (only for USB type)

*: Default setting

3 Operation Mode

MCR-F110 has four operation modes as follows.

Operation mode	Command	Description
Single Reading Mode	SYNCMODE=0	Single reading for each SYNC input.
Timeout Mode	SYNCMODE=1	After SYNC input, the reader continues reading during the duration time set by the ”TOTALLIM” command. The reader sends the error code when it fails in reading within the duration time.
External Trigger Mode	SYNCMODE=2	The code reader continues reading during the SYNC input is ON. The “g” command (start reading command) is not accepted.
Auto-sense Mode	SYNCMODE=3	The reader automatically detects the label, turns on the synchronization input, and performs symbol reading automatically.

The reading operation means that processing from image capture to symbol decoding.

The following diagnostic mode can be used while the reader is in Single Reading Mode.

Diagnostic mode	Command	Description
Normal Mode	TEST=0	For normal use or for returning to Normal Mode.
Test Mode	TEST=1	For measurement of symbol reading rate.
Continuous Reading Mode	continue	Used to adjust the reading position. Exit this mode with the ”stop” command.

The diagnostic modes is used for configuring the code reader. Do not use this mode in normal operation.

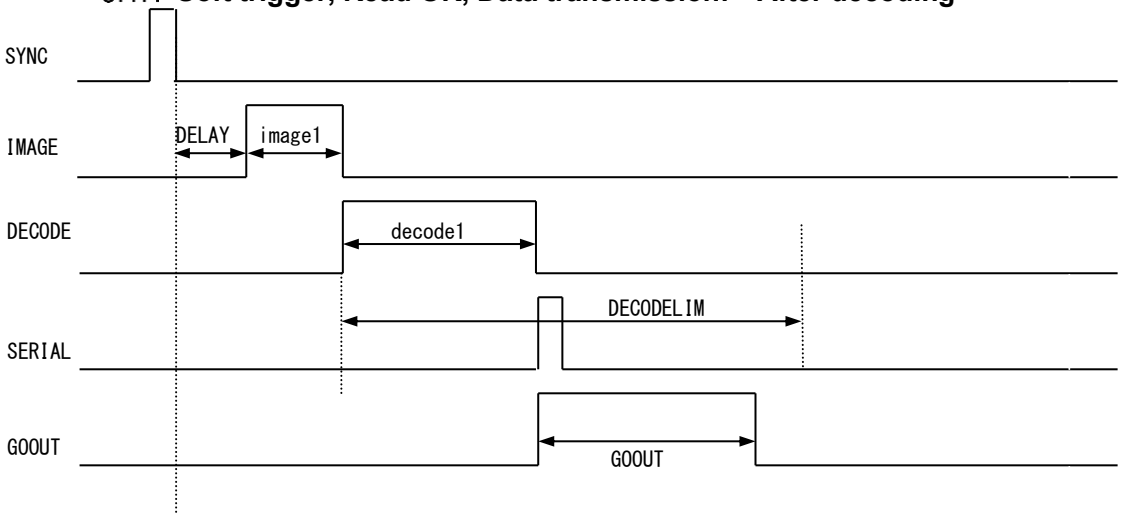
Glossary of terms

DELAY：	The time from SYNC ON to reading.
CHATT：	The time needed for eliminating the chattering.
IMAGE：	The duration for capturing an image.
DECODE：	The duration for decoding.
DECODELIM：	The maximum duration to attempt the decoding.
GOOUT：	The duration of GO signal.
NGOUT：	The duration of NG signal.
SERIAL：	The duration of data output through the serial interface.
MAXIMG：	The maximum number of images in the buffer.
WAITIMG：	The interval of image capture.
Soft trigger：	The SYNC by the serial command input.
Hard trigger：	The SYNC by digital input or the Read key.

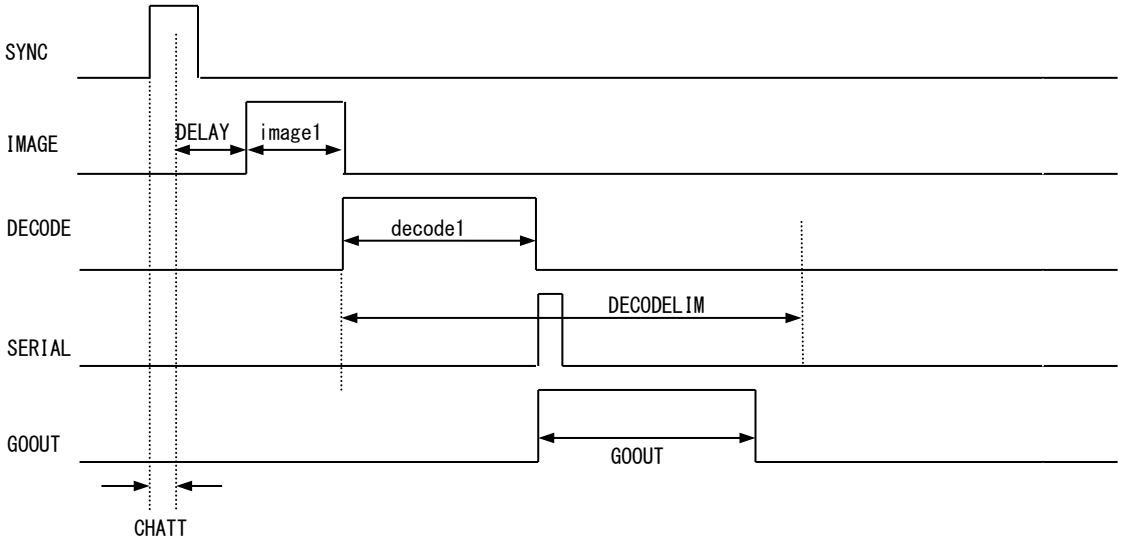
3.1 Single Reading Mode

The code reader performs a single read for each SYNC input.

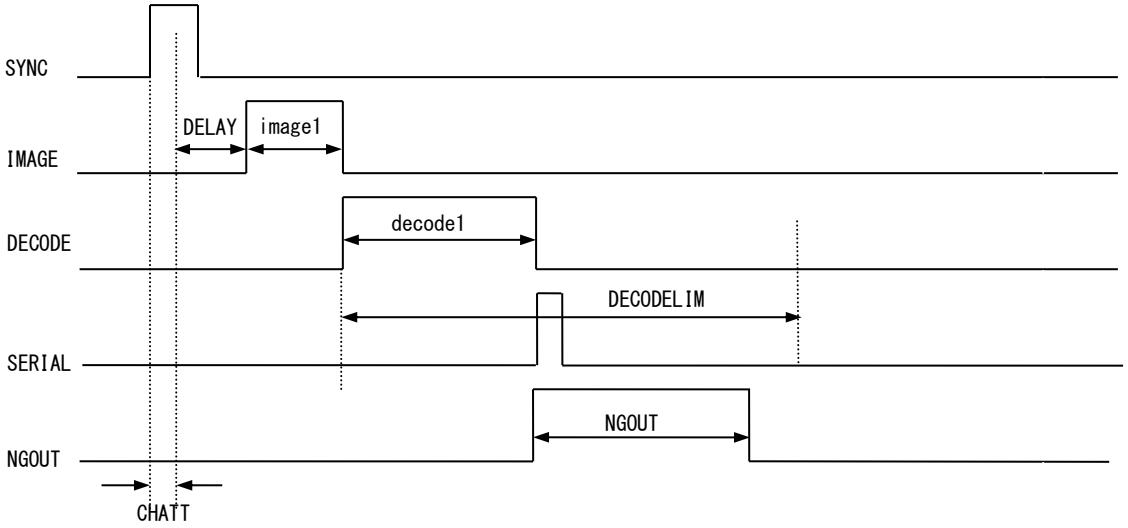
3.1.1 Soft trigger, Read OK, Data transmission: After decoding



3.1.2 **Hard trigger, Read OK, Data transmission:　After decoding**



3.1.3 **Hard trigger, Read NG, Data transmission:　After decoding**



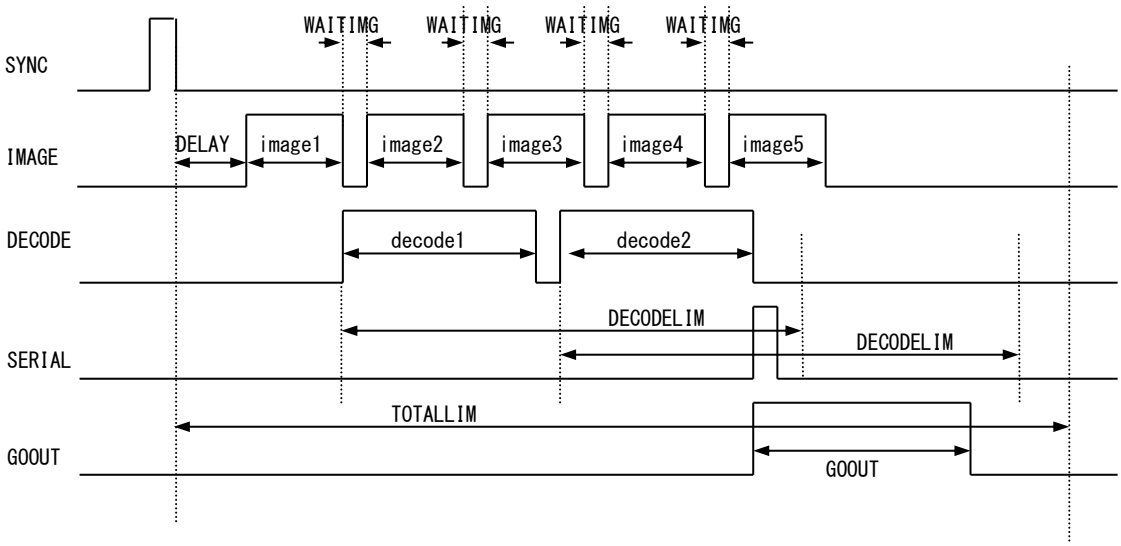
3.2 Timeout Mode

The code reader reads the symbol repeatedly during the duration time, set by ”TOTALLIM” after the SYNC input, or continues reading until the decoding is successful. If the code reader cannot decode successfully in the duration time, it stops reading and sends an error code to the host.

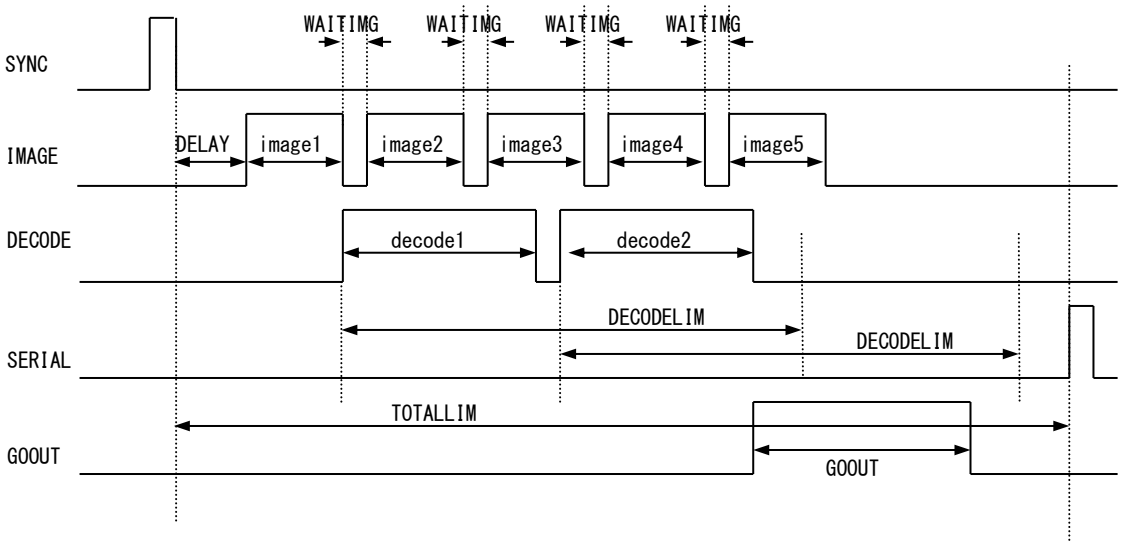
Typically MAXIMG is set to a number greater than one. The multiple MAXIMG enables the Cycle Buffer Function and the parallel processing of the image capturing and decoding. This can reduce the processing time.

The below timing charts explain the behavior of Cycle Buffer Function.

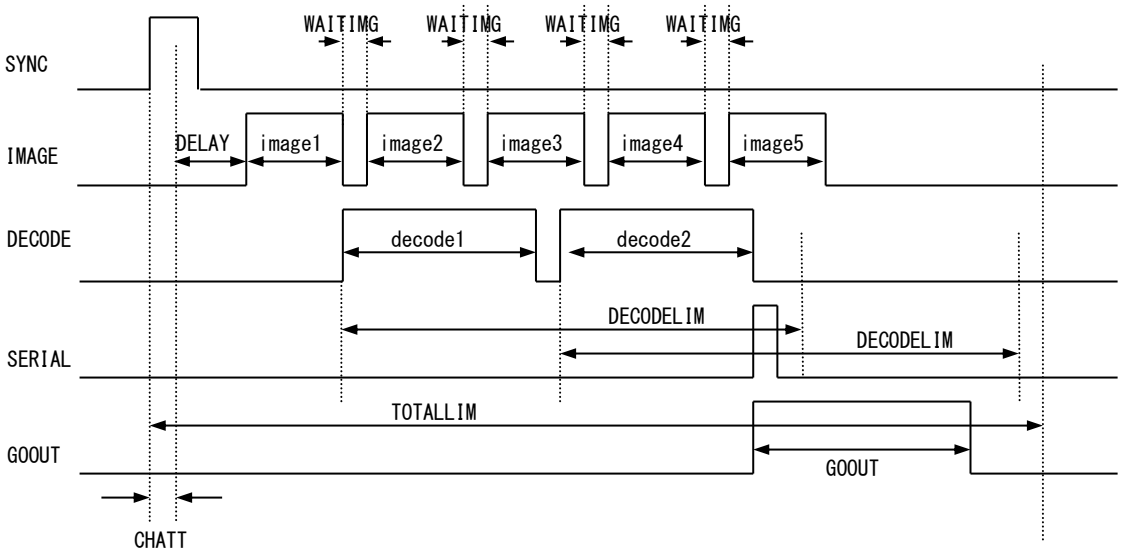
3.2.1 Soft trigger, Read OK, Data transmission: After decoding



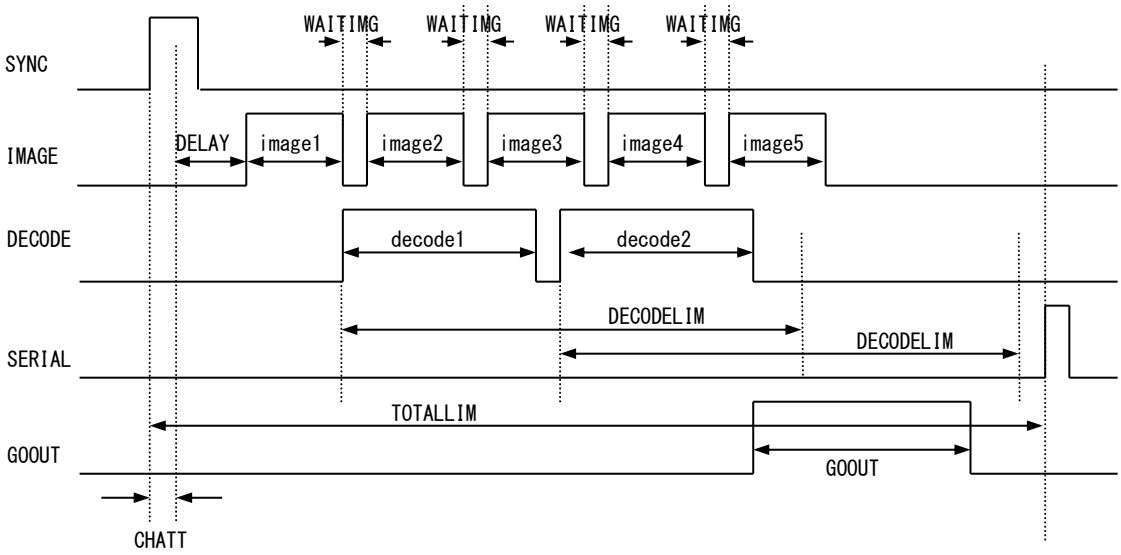
3.2.2 Soft trigger, Read OK, Data transmission: After SYNC OFF



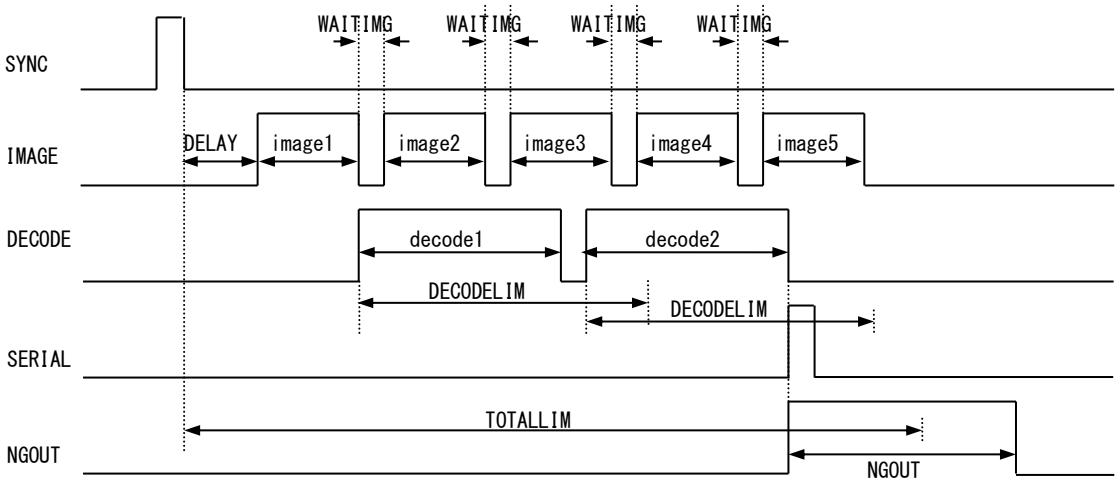
3.2.3 **Hard trigger, Read OK, Data transmission:　After decoding**



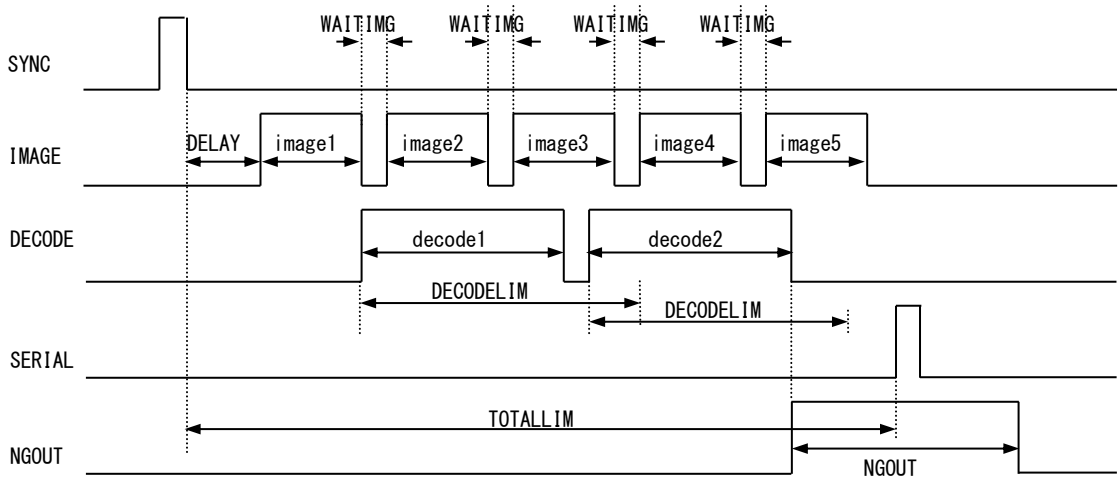
3.2.4 **Hard trigger, Read OK, Data transmission:　After SYNC OFF**



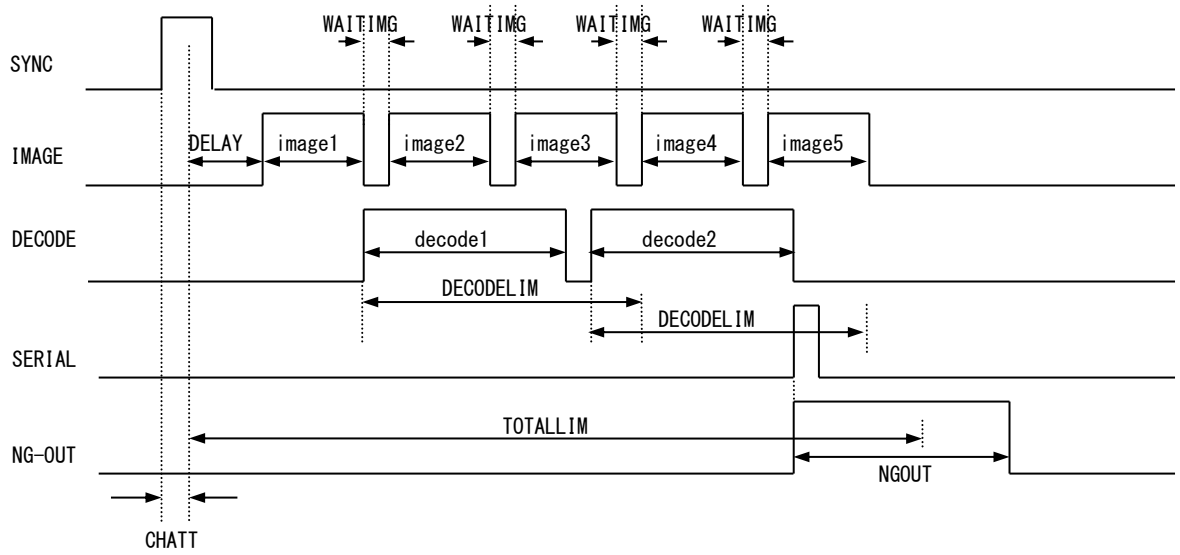
3.2.5 Soft trigger, Read NG, Data transmission: After decoding



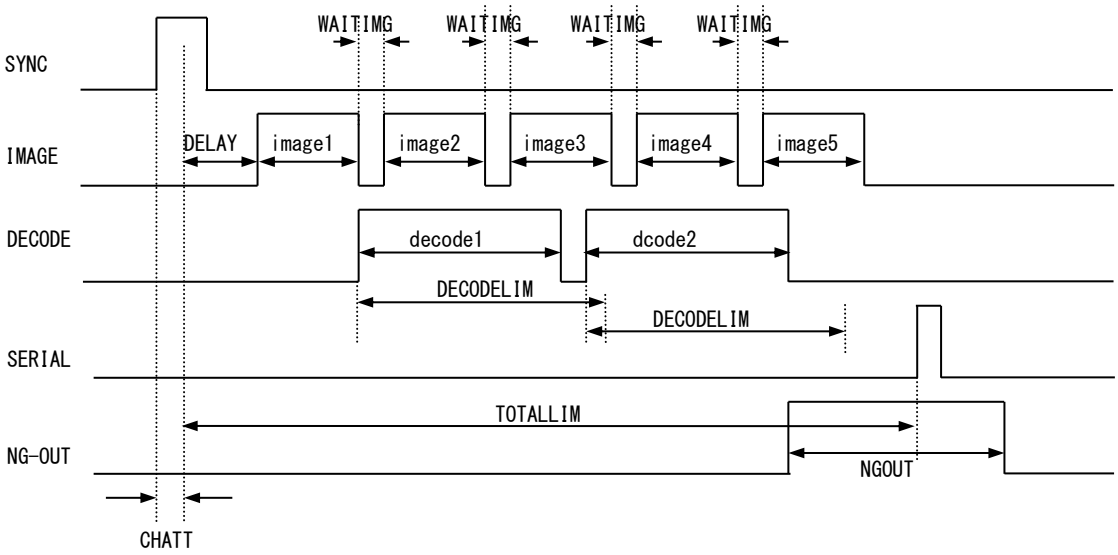
3.2.6 Soft trigger, Read NG, Data transmission: After SYNC OFF



3.2.7 **Hard trigger, Read NG, Data transmission:　After decoding**



3.2.8 **Hard trigger, Read NG, Data transmission:　SYNC OFF**

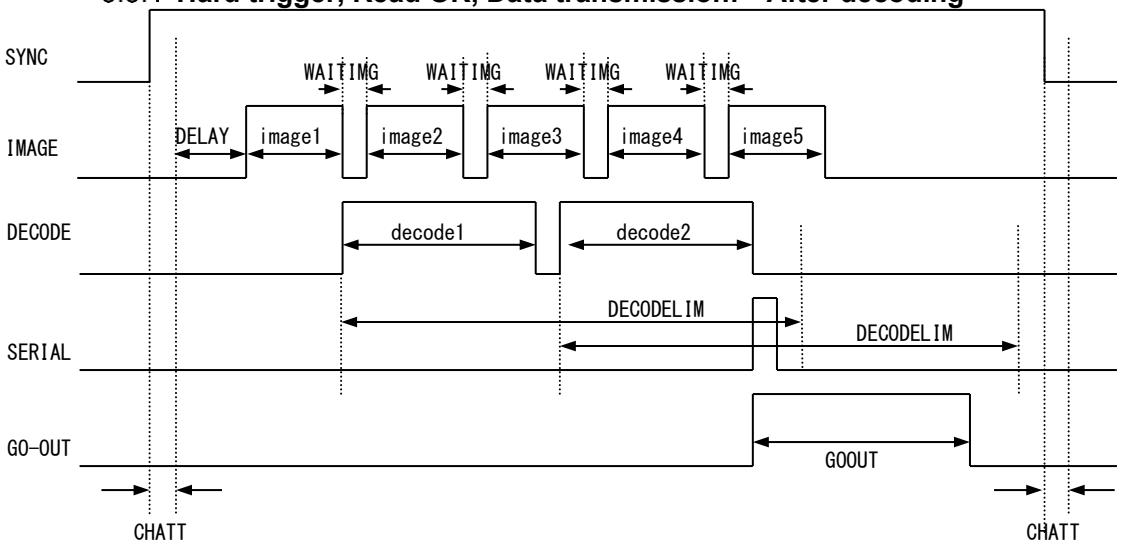


3.3 External Trigger Mode

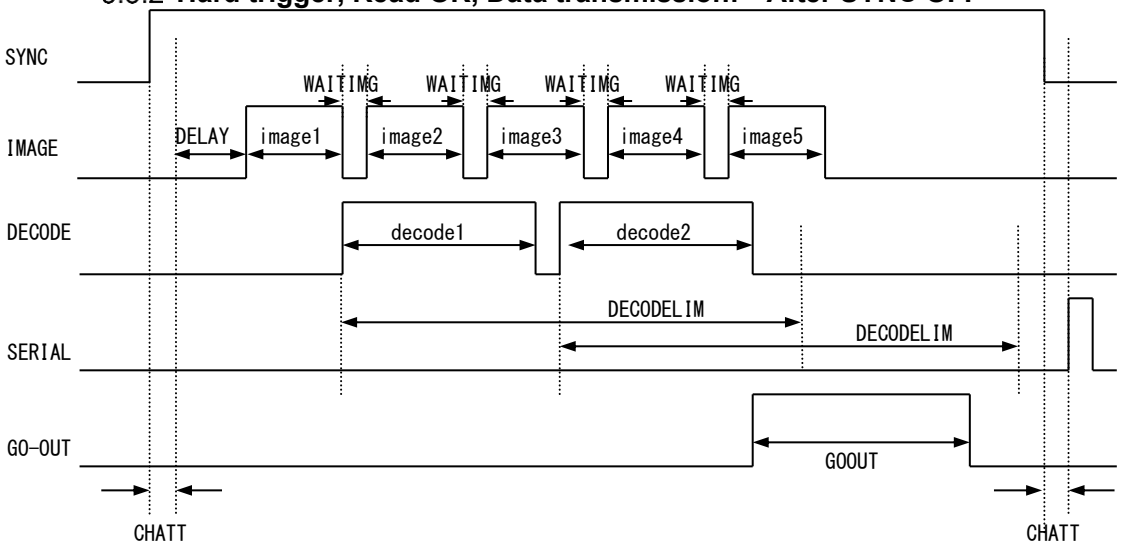
The code reader reads the symbol repeatedly during the SYNC input is active. Typically MAXIMG is set to a number greater than one. The multiple MAXIMG enables the Cycle Buffer Function and the parallel processing of the image capturing and decoding. This can reduce the processing time.

The below timing charts explain the behavior of Cycle Buffer Function.

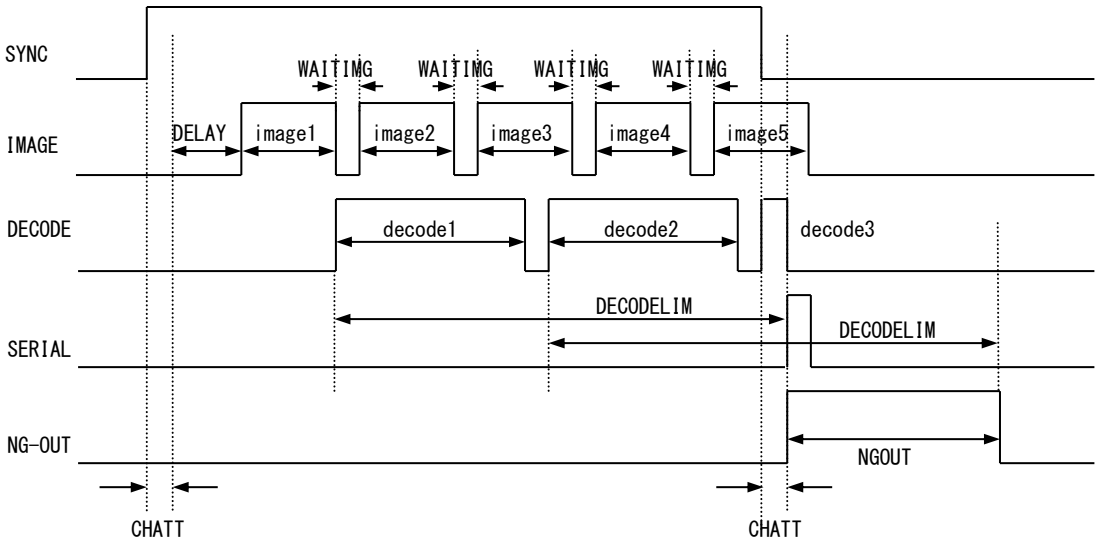
3.3.1 Hard trigger, Read OK, Data transmission: After decoding



3.3.2 Hard trigger, Read OK, Data transmission: After SYNC OFF



3.3.3 Hard trigger, Read NG, Data transmission: After decoding or SYNC OFF



3.4 Continuous Reading Mode

In Continuous Reading Mode, the code reader reads repeatedly. It is useful for camera tunings.

To put the code reader in this mode, send the serial command ”continue” to the code reader, and then send the serial command ”g” to start reading.

To exit this mode, send the serial command ”stop” to the code reader.

It can not be used in Auto-sense Mode.

3.5 Test Mode

In Test Mode, the code reader can measure the reading ratio. It is for checking the reading condition and stability.

To set the code reader in this mode and start, send the serial command ”TEST=1” and “g” to the code reader.

To exit this mode, send the serial command ”TEST=0” to the code reader.

Once this mode is started, the code reader will output an asterisk ”*” to the host for each 10% of the total number of reading (in the default setting, it is set to 10). The code reader will stop the Test Mode after output an asterisk 10 times.

It can not be used in Auto-sense Mode.

(Output example)

```
*** TEST MODE  ***
1 2 3 4 5 6 7 8 9 10
* * * * * * * * * *
NG    1: OK   99/SYNC  100
NG  1.00%: OK 99.00%
```

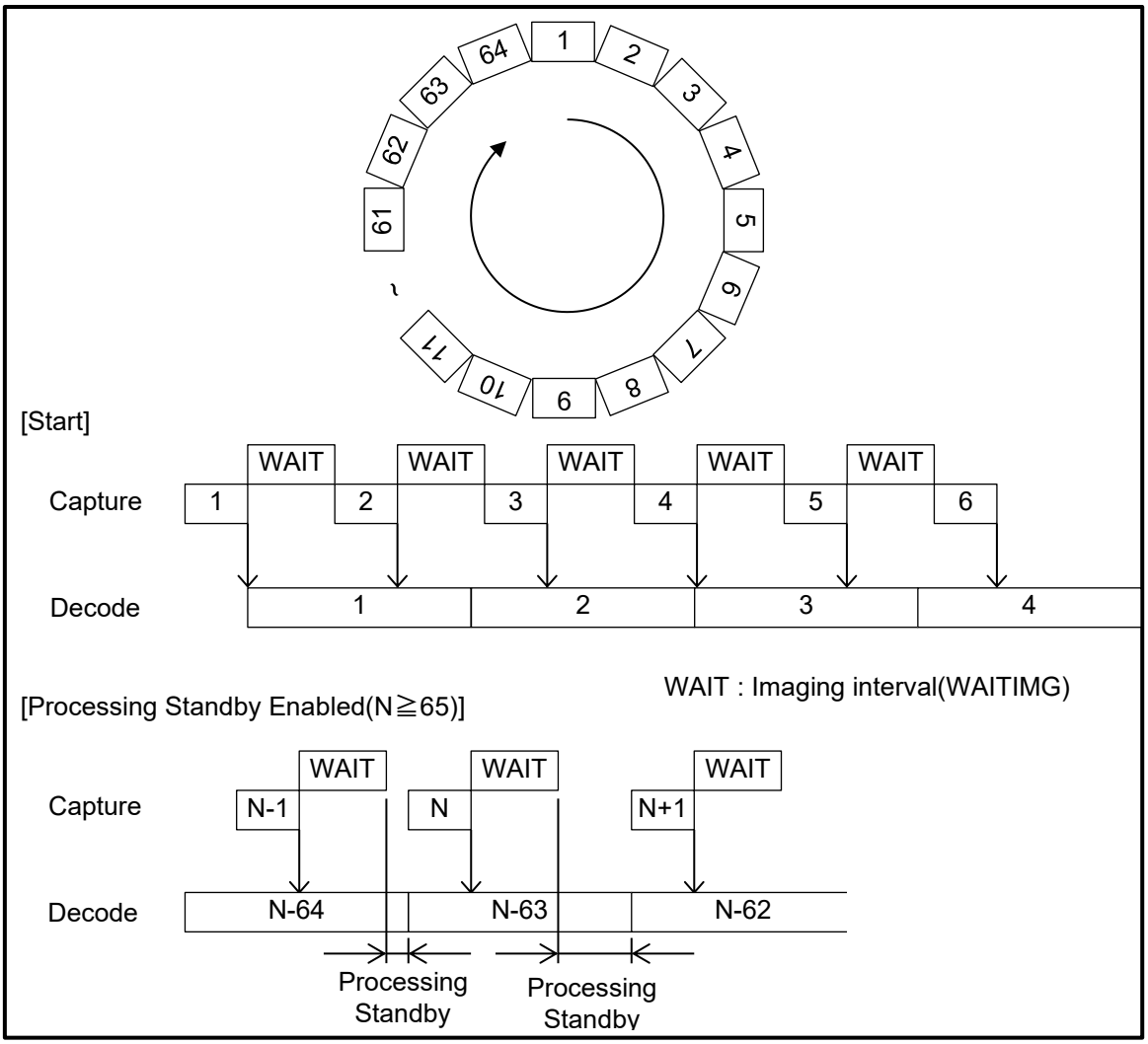
(Output format)

```
NGaaaaa: OKbbbbb/SYNCccccc[CR]
NGddd.dd%: Okeee.ee%[CR]
```

aaaaa :	NG Count	(0 - 1000, maximum number is 1000)
bbbbb :	OK Count	(0 - 1000, maximum number is 1000)
ccccc :	SYNC Input	(NOT displayed more than 1000)
ddd.dd:	NG Rate	(0.00-100.00, Displayed to the second decimal place)
eee.ee:	OK Rate	(0.00-100.00, Displayed to the second decimal place)

3.6 Cycle Buffer Function

Cycle Buffer Function is for moving object, especially for the case it is difficult to catch the object within the field of view. In SYNCMODE=1 (Timeout Mode) or SYNCMODE=2 (External Trigger Mode), the Cycle Buffer Function is enabled when the MAXIMG number is greater than 1 (max. 64). If the Cycle Buffer Function is enabled, the code reader captures multiple images after receiving the SYNC signal and then saves the images to the buffer memory (up to 64 images). The code reader starts decoding from memory #1, if it is not successful, the code reader will decode the image saved in memory #2. When the buffer memory reaches to #64, the next image will be saved into buffer memory location #1 if the code reader has finished decoding that image. If the code reader has not finished, the saving of 65th image will wait until the decoding of memory #1 is finished.



3.7 Concept of Cycle Buffer

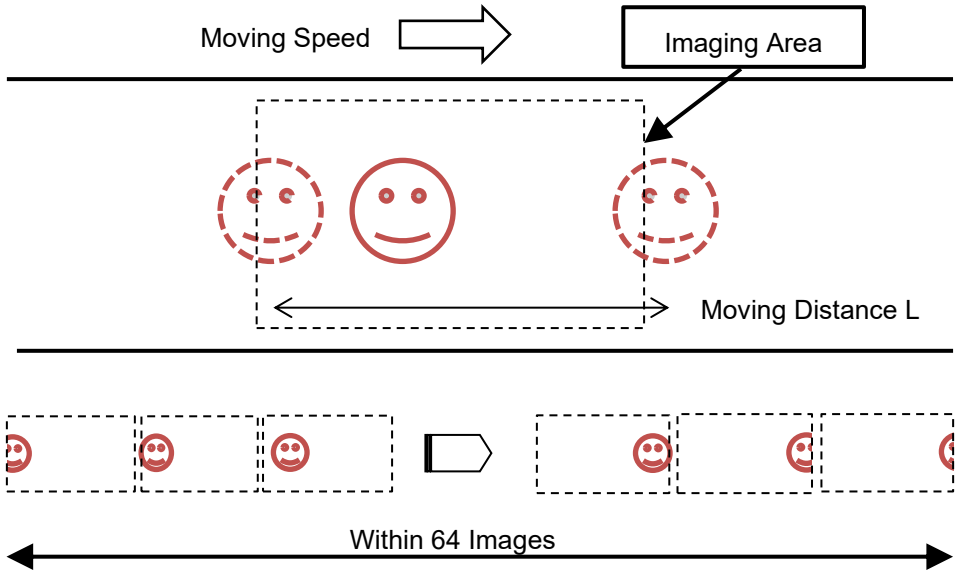
This function focuses to avoid No Read by missing the object from the sight of MCR-F110. Decoding process can be postponed and the reader focuses to obtain images which include the moving object. Please try to configure to finish the capture and decode within the time the reader saves the images whose number is set by MAXIMG. The maximum value of MAXIMG is 64. Otherwise, the imaging interval will vary depending on the reading processing time after the number of imaging.
The less capture resolution (number of pixels), the less capture time and it results to shorten the imaging interval.

[Configuration Example]

The relation of Shutter Time (SHUTT), Capture Time (CAPIMG), Capture Interval (WAITIMG) and Total Imaging Cycle (CAPCYC) are as follows.

CAPCYC ＝ SHUTT + CAPIMG + WAITIMG

Moving Time T = L / V should be less than (64 × CAPCYC)
when the moving object's speed is V and the object's moving distance is L.



MAXIMG=64、SHUTT=1/1000s=1ms、CAPIMG=20ms、WAITIMG=10ms
Imaging Cycle is CAPCYC=SHUTT+CAPIMG+WAITIMG=1ms+20ms+10ms=31ms.
(1)Moving speed V when the imaging range is L=100mm is:
L/V < 64×CAPCYC
−> V > L/(64×CAPCYC) = 100mm/(64×31ms) = 0.05m/s = 3.0(m/min.) ∴ V > 3.0m/min.
(2)Imaging range L when the moving speed is V=30m/min.(= 500mm/s = 0.5mm/ms) is:
L/V < 64(CAPCYC)
−> L < 64×CAPCYC×V = 64×31ms×0.5mm/ms = 992mm ∴ L < 992mm

4 Configuration for symbol reading

4.1 Automatic parameter tuning (Auto-Tuning)

With this function, the code reader can be configured automatically. This function is available by the communication from the host device as well as an easy key operation on the code reader.

For the detail of the key operation, please refer to 4.1.3 “Auto-Tuning by key operation”.

4.1.1 How to operate Auto-Tuning function

Execute the Auto-Tuning by MCRConfig or serial command.

- Place and fix the symbol within the reading distance range.
- Follow the steps below.

1. Start Auto-Tuning by MCRConfig or serial command.
The monitor LED status: OFF

↓

2. Wait until Auto-Tuning is completed.
The monitor LED status: Orange ■ | Blinks

↓

3. The result of Auto-Tuning
 - Success The monitor LED status: Green ■ | ON
 - Failure The monitor LED status: Red ■ | Blinks 7 times at short intervals

↓

4. Save the settings when Auto-Tuning is successful.
(1) Save one type of settings in specified table:
The monitor LED status: Green ■ | Blinks 3 times
(2) Save up to 3 types of settings in Table #1 to #3:
The monitor LED status: Green ■ | Blinks 4 times

Note: If there is no operation for 40 seconds, it will return to the standby state without saving the result.



DO NOT turn off power during Auto-Tuning operation.
DO NOT send serial commands during Auto-Tuning operation.

4.1.2 **Commands for Auto-Tuning**

In Auto-Tuning, you can configure the Auto-Tuning operation, reference of tuning result, selection of storing method and setting of tuning condition.

- Start Auto-Tuning
  *setup
  This command is to start this function.

- Force-quit Auto-Tuning
  *cancel
  This function takes relatively long time. The code reader will force-quit by this command.

- Check the reader state
  *?state
  This command is to check the state of the reader.

State	Response
Reading standby	STANDBY[CR]
Auto-Tuning in progress	RECEIVED[CR]
Auto-Tuning success: waiting for save settings	SAVEWAIT[CR]
Auto-Tuning failure	ABORTED[CR]
Read in progress	(No reply)

- Save the result of Auto-Tuning (Up to 3 types)
  *save
  This command is to save the results. When the reader receives this command within about 40 seconds after the completion of the tuning process, the results are saved in Table #1 to #3. If no command is received, the reader will discard the results and return to reading standby state.

- Save the result of Auto-Tuning (1 type)
  *save2=a (a: Table number #1 - #16)
  This command is to save one type of results. When the reader received within about 40 seconds after the completion of the tuning process, the results are saved in Table ”a”. If no command is received, the reader will discard the results and return to reading standby state.
  If no table is specified, the results are saved in Table #1.

- Refer to the saved result
  ?ddmtbl
  This command is to refer to the saved result of Auto-Tuning.

Output example of ?ddmtbl

```
************ STATUS ************
 last table number=0(read only)
 NEXTDDMTBL=1 (1-16)
 DDMTBLNG=1 (0:OFF 1:ON)
 ENABLEDDMTBL=1 (1-16)
 EDITDDMTBL=1 (1-16)
 TBLTX=0 (0:OFF 1:ON)
 +++++ table 0 +++++
 DDMtblname=table0
 DDMcapmode=0,0,752,480
 DDMflash=100,100
 DDMshutt=3
 DDMagain=9
 DDMdgain=4
 DDMblackval=64
 DDMdecodelim=500
 DDMlight=3
 DDMmirror=2
 DDMpreproc=0,0,0,0,0
 +++++ table 1 +++++
 DDMtblname=table1
 DDMcapmode=0,0,752,480
 DDMflash=100,100
 DDMshutt=3
 DDMagain=9
 DDMdgain=4
 DDMblackval=64
 DDMdecodelim=500
 DDMlight=3
 DDMmirror=2
 DDMpreproc=0,0,0,0,0
 +++++ table 1 +++++
  (omission)
 +++++ table 16 +++++
 DDMtblname=name
 DDMcapmode=0,0,752,480
 DDMflash=100,100
 DDMshutt=2
 DDMagain=9
 DDMdgain=1
 DDMblackval=128
 DDMdecodelim=500
 DDMlight=3
 DDMmirror=2
 DDMpreproc=0,0,0,0,0
*************** END ***************
MCR-F110U : 077-V1.1a
```

4.1.3 **Auto-Tuning by key operation**

Execute Auto-Tuning by key operation of the reader.

- Place and fix the symbol within the reading distance range.
- Follow the steps below.

1. Start Auto-Tuning by holding down the Teach-key.
The monitor LED status: Orange ■ | ON

↓

2. Wait until Auto-Tuning is completed.
The monitor LED status: Orange ■ | Blinks

↓

3. The result of Auto-Tuning
 - Success The monitor LED status: Green ■ | ON
 - Failure The monitor LED status: Red ■ | Blinks 7 times at short intervals

↓

4. Save the settings when Auto-Tuning is successful.
(1) Save one type of settings in specified table:
 (1-1) Holding down the Read key
  The monitor LED status: Red ■ | ON
 (1-2) Save successful
  The monitor LED status: Green ■ | Blinks 3 times

(2) Save up to 3 types of settings in Table #1 to #3:
 (2-1) Holding down the Teach key
  The monitor LED status: Red ■ | ON
 (2-2) Save successful
  The monitor LED status: Green ■ | Blinks 4 times

Note: If there is no operation for 40 seconds, it will return to the standby state without saving the result.



DO NOT turn off power during Auto-Tuning operation.
DO NOT send serial commands during Auto-Tuning operation.

4.2 Table Mode [AGC=T]

MCR-F110 works in Table Mode, which has up to 16 combinations of camera settings (hereinafter, referred to as “tables”) and uses each table in a sequence of the reading.

Switch to the table mode with the command “AGC=T”.
It can not be used together with Auto Gain Control Mode.

The number of tables to be used is set by the command below.

ENABLEDDMTBL=a (a: 1 to 16)

e.g.1) Use 3 tables (a=3)
Table #1, #2 and #3 are enabled and Table #4 to #16 will not be used. If the code reader successfully decodes a symbol with Table #1, Table #1 will be used in the next reading.
If the code reader failed to decode a symbol with Table #1, the settings will switch to Table #2, and if it fails again, switch to Table #3.
If the code reader failed to decode with Table #3, the settings will switch back to Table #1.

e.g.2) Use 1 table (a=1)
Only Table #1 is enabled. Table #2 to #16 will not be used. As the result, Table #1 is always used every reading.

Set the number of tables according to the reading target and the operating environment.

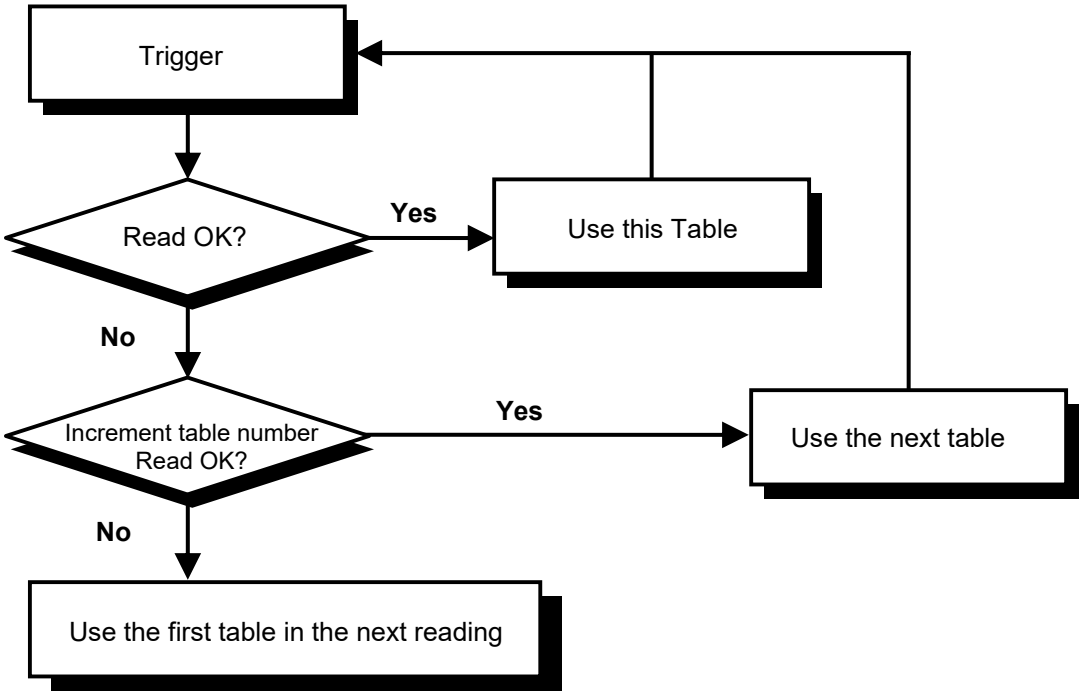
The following shows the structure of the Table. Each table has 11 types of parameters:

(1)	Reading Area	[DDMcapmode]
(2)	Illumination Control (ON/OFF)	[DDMlight]
(3)	Illumination brightness	[DDMflash]
(4)	Shutter Speed	[DDMshutt]
(5)	Analog Gain	[DDMagain]
(6)	Digital Gain	[DDMdgain]
(7)	Black Level Illumination	[DDMblackval]
(8)	Image Preprocessing	[DDMpreproc]
(9)	Mirrored Image	[DDMmirror]
(10)	Decode Time Limit	[DDMdecodelim]
(11)	Table Name	[DDMtblname]

When the reading is successful, the code reader uses the table settings for the next reading.
When the reading is unsuccessful, the code reader attempts to read using the next table settings.

The flow chart of Table mode is shown below.

< Flow char of Table Mode >



4.2.1 **Edit table**

Edit tables with serial commands.

(1) Set the target Table #.
Send the serial command below to assign the table number to be edited.
EDITDDMTBL=a (a: 1 to 16)

(2) Configure the parameters.
Refer to section 4.3 “Setting parameters for Table Mode” for detailed information.

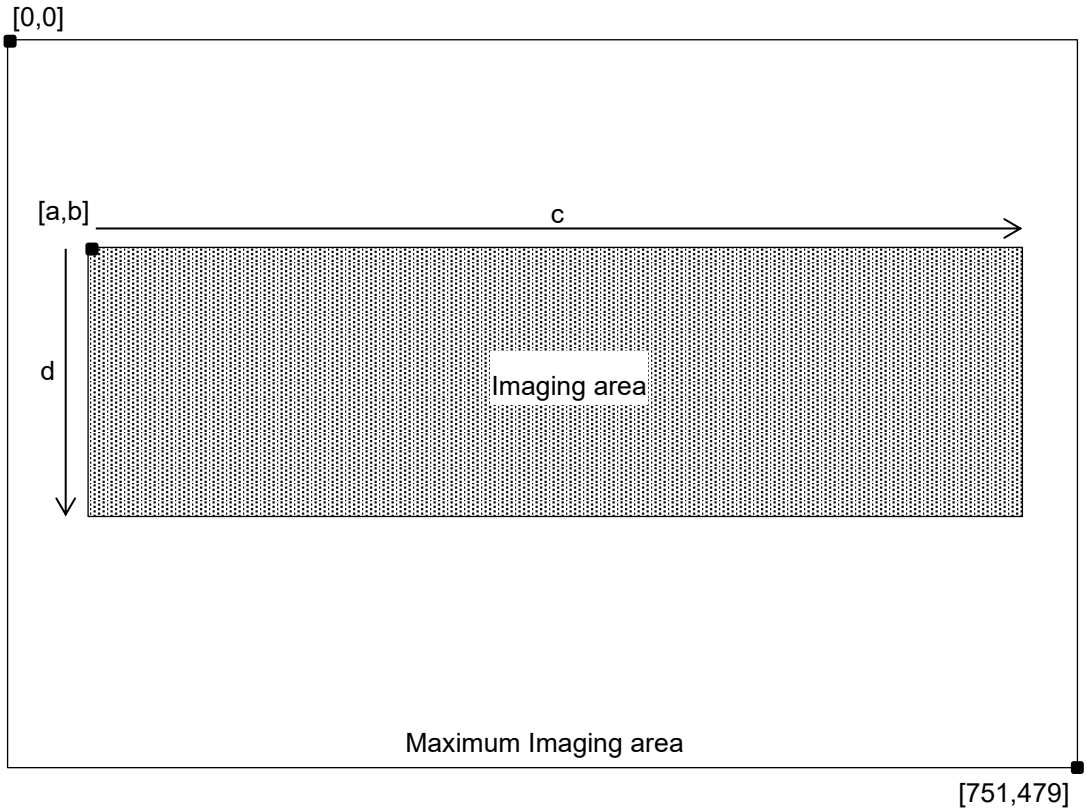
(3) Edit another table
Repeat the step 1 & 2.

(4) Check the target Table
You can check the target table status using the following command.
?ddmtbl

4.3 Setting parameters for Table Mode

4.3.1 Reading area [DDMcapmode]

This command specifies the reading area from an captured image.
The parameters are original coordinate and value of width (x) & height (y).



DDMcapmode=a,b,c,d

a: Coordinate origin X (0 to 742)　　b: Coordinate origin Y (0 to 470)

c: Width (10 to 752)　　d: Height (10 to 480)

- Specify the full screen

  DDMcapmode=0,0,752,480

- Specify the 50 % of the center

  DDMcapmode=188,120,376,240

† Configuring the reading area too narrow may deviate the symbol from the reading area and may cause a read error. Make sure that the size of the reading area fits the symbol.

4.3.2 **Illumination control (ON/OFF) [DDMlight]**

This command specifies the behavior of internal illumination.

DDMlight=a

0:	OFF
3:	ON when capturing an image
10:	Always ON
11:	ON during reading (SYMCMODE=1 or 2 only)

4.3.3 **Illumination brightness [DDMflash]**

This command specifies the brightness of each illumination.

DDMflash=a,b

a (Upside illumination), b (Downside illumination)

0 : OFF

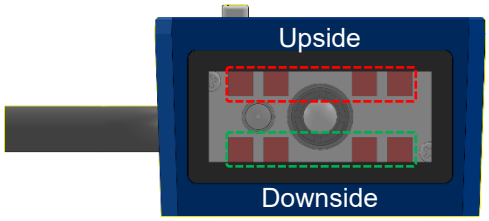
1 (dark) to 100 (bright): ON

To set to ON, set in the range of 1 to 100, but you can set only the same value for a and b. You can not set different values.

If “b” is omitted, the value of “a” will be set.

e.g.) ○: Settable　×: Not settable

DDMflash=75,75	○: Both upside and downside ON (brightness 75)
DDMflash=75	○: Both upside and downside ON (brightness 75)
DDMflash=75,0	○: Upside ON (brightness 75), downside OFF
DDMflash=0,75	○: Upside OFF, downside ON (brightness 75)
DDMflash=75,25	×: Upside ON (brightness 75), downside ON (brightness 25)



4.3.4 Shutter speed [DDMshutt]

This command specifies the shutter speed of the reader. A slower shutter speed makes the image brighter but may introduce motion blur into the image.

DDMshutt=a

0: 1/60	1: 1/125	
2: 1/250	3: 1/500,	
4: 1/1000	5: 1/2000	
6: 1/4000	7: 1/6000,	
8: 1/8000	9: 1/10000	(Unit: seconds)

4.3.5 Analog gain [DDMagain]

This command specifies the analog gain of the camera.
A larger gain makes the image brighter.

DDMagain=a (a: 1 to 25, 1: Lowest gain, 25: Highest gain)

4.3.6 Digital gain [DDMdgain]

This command specified the digital gain of the camera.
A larger gain makes the image brighter.

DDMdgain=a (a: 1 to 12, 1: Lowest gain, 12: Highest gain)

4.3.7 Black level adjustment [DDMblackval]

This command is used to improve the contrast of an image taken by the reader. It may be suitable for lower contrast images.

DDMblackval=a (a: 1 to 255, 1: Black emphasis, 255: white emphasis)

4.3.8 Image preprocessing [DDMpreproc]

If the image quality is low, by using an appropriate image filter may improve the reading ratio.

4.3.9 Mirror image [DDMmirror]

This command enable / disable mirrored image symbol.

DDMmirror=a

0: Read only normal type (not mirrored)
1: Read only mirrored type
2: Read both of normal type and mirrored type

4.3.10 Decode time limit [DDMdecodelim]

This command specifies the decode time limit.

DDMdecodelim = a (a: 50 to 10000, Unit: ms)

4.3.11 Table name [DDMtblname]

This command specifies a name of the current table number. This command is just for a memo. It does not affect any performances of the reader.

DDMtblname=a (a: Up to 16 one-byte characters)

4.4 Auto Gain Control Mode [AGC=E]

In Auto Gain Control Mode, the reader automatically adjusts the parameter (shutter speed, analog gain and digital gain) settings to the ambient brightness.

Switch to the table mode with the command “AGC=E”.
It can not be used together with the Table mode.

4.5 Reading parameters for Auto Gain Control Mode

4.5.1 Reading area [CAPMODE]

CAPMODE=a,b,c,d
Refer to 4.3.1 ”Reading area [DDMcapmode]” for details.

4.5.2 Illumination control (ON/OFF) [LIGHT]

LIGHT=a
Refer to 4.3.2 “Illumination control (ON/OFF) [DDMlight]” for details.

4.5.3 Illumination brightness [FLASH]

FLASH=a,b
Refer to 4.3.3 “Illumination brightness [DDMflash]” for details.

4.5.4 Shutter speed [SHUTT]

SHUTT=a
Refer to 4.3.4 “Shutter speed [DDMshutt]” for details.

4.5.5 Analog gain [AGAIN]

AGAIN=a
Refer to 4.3.5 “Analog gain [DDMagain]” for details.

4.5.6 Digital gain [DGAIN]

DGAIN=a
Refer to 4.3.6 “Digital gain [DDMdgain]” for details.

4.5.7 Black level adjustment [BLACKVAL]

BLACKVAL=a
Refer to 4.3.7 “Black level adjustment [DDMblackval]” for details.

4.5.8 Image preprocessing [IPFUNC0]

IPFUNC0=a,b,c,d,e
Refer to 4.3.8 “Image preprocessing [DDMpreproc]” for details.

4.5.9 **Mirror image [MIRROR]**

MIRROR=a

Refer to 4.3.9 “Mirror image [DDMmirror]” for details.

4.5.10 **Decode time limit [DECODELIM]**

DECODELIM=a

Refer to 4.3.10 “Decode time limit [DDMdecodelim]” for details.

5 Auto-sense Mode

In Auto-sense mode, the reader automatically detects and reads the symbol. It detects the change in ambient brightness and the movement of the subject, etc., and performs scanning automatically. This function is useful for reading documents with symbols printed on them in front of the reader.



- The Auto-sense mode works with the brightness of normal room lighting, but may not be suitable for reading under extremely bright and dark environments such as outdoors, or under flickering lighting.
- In Auto-sense mode, other operation modes do not work.
  *: Refer to 3 “Operation Mode”.
- In Auto-sense mode, the operation of the Teach key and the Read key are invalid.

5.1 How to use

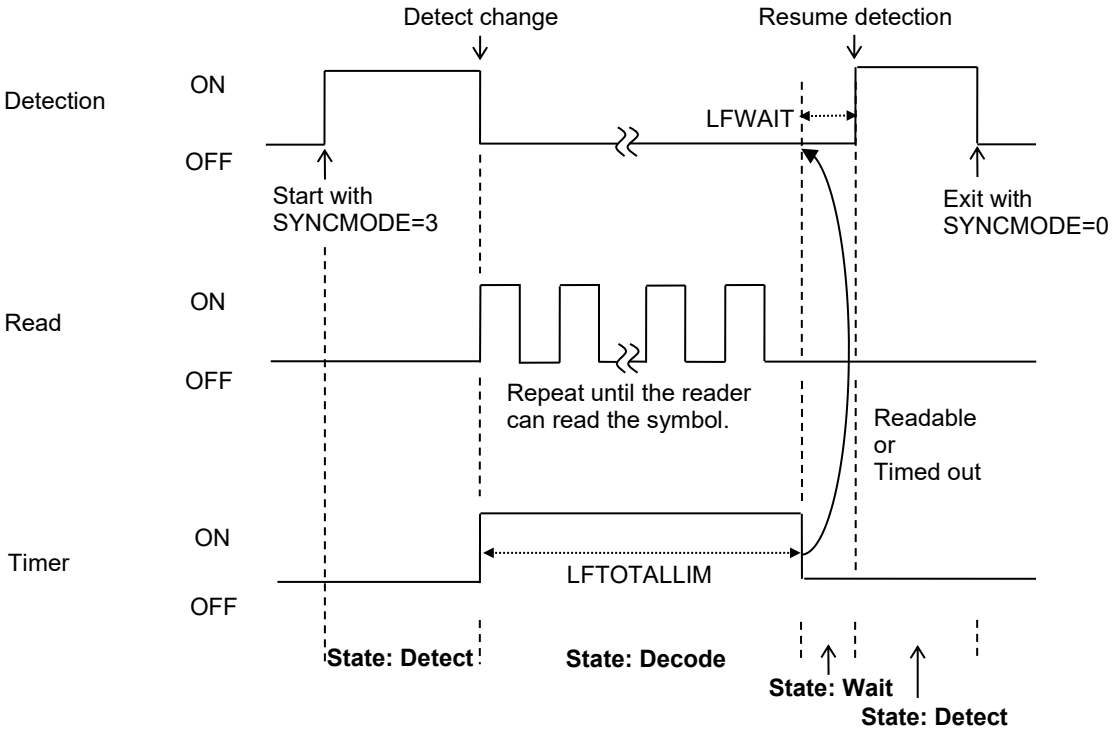
- Start Auto-sense Mode
  - Send the serial command “SYNCMODE=3” to the reader.
  - Enable the Auto-sense Mode by reading the setting code. Refer to the 2D code menu.

  However, it can not start in the following cases.
  - When set to continuous reading mode
  - When set to test mode
  - When running the Auto-Tuning function
  - When synchronous input is enabled in PLC-link

- Exit Auto-sense Mode
  - Send the serial command “SYNCMODE=0” to the reader.
  - Disable the Auto-sense Mode by reading the setting code. Refer to the 2D code menu.

5.2 Timing chart

While in Auto-sense Mode, the reader transitions in three state: detect, decode and wait.



5.3 Setting parameters for Auto-sense Mode

5.3.1 Pointer, Illumination [LFILLUM]

This command specifies the pointer and illumination during detection.

LFILLUM=a (a: 0 to 2)

0: Pointer OFF, Illumination OFF

1: Pointer ON, Illumination ON

2: Pointer OFF, Illumination ON

5.3.2 Detection sensitivity [LFSENS]

This command specifies the detection sensitivity. If the sensitivity is too good, lower the value; if not good, increase the value.

LFSENS=a (a: 0 to 4, 0: High sensitivity, 4: Low sensitivity)

5.3.3 Reading timeout [LFTOTALLIM]

After detection, the reading operation is performed until the symbol reading succeeds or the timeout occurs. Set this time.

LFTOTALLIM=a (a: 0 to 10, Unit: seconds)

5.3.4 Wait time for resuming detection [LFWAIT]

The reader waits for a fixed time and resumes processing after a successful symbol read or read timeout. Set this wait time.

LFWAIT=a (a: 0 to 10, Unit: seconds)

5.3.5 Continuous reading of the same symbol [VERIFY]

In Continuous Reading Mode or Auto-sense Mode, this command specifies whether to permit continuous reading of the same symbol.

VERIFY=a (a: 0: Permission, 1: Prohibition)

5.3.6 Wait time for continuous reading of the same symbol [VWAIT]

This command specifies the prohibited time when VERIFY=1.

VWAIT=a (a: 1 to 20, Unit: x 500ms)

5.3.7 Status of auto-sense mode [?LF]

?LF

（Output example）

```
************* STATUS *************
 LFILLUM=1 (0:off 1:pointer 2:flash)
 LFSENS=2 (0-4)
 LFTOTALLIM=3 (1-10[sec])
 LFWAIT=1 (0-10[sec])
 VERIFY=0 (0:off 1:on)
 VWAIT=3 (1-20 x500[ms])
*************** END ***************
MCR-F110U : 077-V1.1a
```

6 Advanced Features

6.1 Preset Mode

In Preset Mode, MCR-F110 reads only the symbols whose data match the preset data. This function has two modes, their registration methods and data collation methods of preset data are different.

```
PREM=0[CR](*):  Preset Mode 0 (Preset Mode is disabled)
PREM=1[CR]:     Preset Mode 1 (Preset data is the first decoded one after power-up)
PREM=2[CR]:     Preset Mode 2 (Preset data is registered in advance)
```

*: Default

To disable Preset Mode, send the command ”PREM=0”.

6.1.1 Preset Mode 1

In this mode, the preset data is the first decoded data after power up. Data comparison is only successful when the code matches completely. The preset data must be set as every time the code reader is powered on because the data is not saved to internal flash memory.

Setup procedure of Preset Mode 1:

1. Send the following commands

   ```
   PREM=1[CR]
   WSETS[CR]:        Save the settings to internal flash memory.
   ```

2. Power-on reset
3. Read the symbol that contains the data to be used as the preset data.

6.1.2 Preset Mode 2

In this mode, the preset data is configured by the serial command. Data comparison is completed by both complete and partial matching. It is possible to save the preset data to internal flash memory.

Send the following command to set this mode:

```
PREM=2[CR]
SET=PREDabcd[CR]:      Set the preset data (data is ” abcd” )
SET=PRENa[CR]:         Set the number of digits (a: 0 to 100)
WSETS[CR]:             Save the settings to internal flash memory
```

e.g.1) Example 1: Only ”12345” is valid

12345: match
1234: mismatch
123456: mismatch
0**12345**: mismatch

```
PREM=2[CR]
SET=PRED12345[CR]
SET=PREN5[CR]
WSETS[CR]
```

e.g.2) Example 2: Valid when first 4 digits are ”ABCD”.

ABCD: match
ABCD333: match
ABCD777777: match
ABC: mismatch
DABC333: mismatch
ABC777777: mismatch

```
PREM=2[CR]
SET=PREDABCD[CR]
SET=PREN0[CR]
WSETS[CR]
```

e.g.3) Example 3: Valid when the length is 10 digits and the data from 3rd-digits to 6th-digits are ”ALFA”.

00**ALFA**1234: match
AA**ALFA**AAAA: match
00**ALFA**12345: mismatch
0A**LFA1**2345: mismatch

```
PREM=2[CR]
SET=PRED??ALFA????[CR]
SET=PREN10[CR]
WSETS[CR]
```

6.1.3 **Output the status of Preset Mode**

?pre[CR]　　: Output the preset status

(Output example)

PREM=0 (0: non 1: power on 2: saved)
PRESET LENGTH: 13
DATA(HEX):
34 39 3F 3F 3F 3F 3F 3F 3F 3F
3F 3F 3F
DATA(ASCII):
49???????????

(Output format)

PREM=a (0: non 1: power on 2: saved)[CR]
PRESET LENGTH: b[CR]
DATA(HEX): [CR]
XX XX XX XX XX XX XX XX XX XX[CR]
XX XX XX XX XX XX XX XX XX XX[CR]
XX XX XX XX [CR]
DATA(ASCII): [CR]
xxxxxxxxxxxxxxxxxxxxxxx

a　　: Preset Mode
b　　: The digit of the length of the preset data
XX XX …　　: Preset data character (in hex)
xxxxxx…　　: Preset data character (in ASCII/JIS)

If the preset data include control code, it is transferred to ”*”.

6.2 Output Additional Information

6.2.1 Warning Information to avoid Read NG

Add warning information that can monitor the reading condition.

Output format: (CcccEeeeDddd)
ccc: Contrast information
eee: Unused error correction rate
ddd: Decodability margin

e.g.1) Read OK
Data Output: (C082E100D091)A123456A
The code reader outputs the contrast value of a symbol including quiet zone. The decoder determines a rectangular area for the symbol and calculates the contrast value from its brightness level. If there are multiple symbols, the calculated value is for the first decoded symbol.

e.g.2) Read NG
Data Output: (C000E000D000)BR
”000” is added in case of Read NG.

6.2.2 Symbol Type Information

Add symbol type information (data carrier identifiers) to the head of decoded symbol and output is in serial. Symbol type is not added at read NG.

Symbol Type	Example of Data Carrier Identifiers
Data Matrix	]d1
QR Code	]Q1
PDF417	]L0
Aztec Code	]z0
Code39	]A0
Code128	]C0
GS1-128	]C1
Codabar	]F0
Code93	]G0
ITF(Interleaved 2 of 5)	]I0
EAN/JAN 13 digits	]E0
EAN/JAN 8 digits	]E4
UPC	]E0
GS1 DataBar	]e0
Composite	]e1

6.2.3 **Decoding Time**

Add decoding time to the symbol data.

6.2.4 **Total Reading Time Information**

Add the number of images captured after trigger input and total time from trigger input to the serial data output. Total reading time is not added at read NG.

Output format: (aa, bbbbbms)

aa: Number of images, bbbbb: Total time [ms]

6.2.5 **Contrast Information**

Add the Contrast Information of the last decoded image.

- Read OK

  Calculate the contrast value of a symbol including a quiet zone. The decoder determines a rectangular area for the symbol and calculates the contrast value from its brightness level.
  If there are multiple symbols, the calculated value is for the first decoded symbol.

- Read NG

  Calculate the contrast value for the whole image.

Output format: (aaa, bbb, ccc) 13 bytes

aaa : Maximum value of reflectance	(000 to 255)
bbb : Minimum value of reflectance	(000 to 255)
ccc : Contrast [%] =100 x (aaa - bbb) / 255	(000 to 100)

6.2.6 **Quality Information**

The percentage of the Unused Error Correction codeword of the 2D code is added to the decoded data.
For example, if the decoder does not use any error correction, this value would be 100. The code reader outputs the quality information individually if there are multiple symbols. Quality Information is not added at Reading Error.

Output format: (QT: ddd)　　8 bytes
ddd: Quality (000 to 100)

6.2.7 **Symbol Coordinates**

The positional data of the symbol in the image is output along with the decoded data. There are two types of output information; barycentric coordinates and rectangular coordinates.

(1) Barycentric coordinates

The origin (+0000, +0000) is the left-top edge of a field of view.

(Example output)

10000000099070(+0122,+0024)

(Output format)

Header + Data + (abbbb,acccc) + Terminator
a : + or -
bbbb: X coordinate of a symbol's barycenter (9999 to +9999)
cccc: Y coordinate of a symbol's barycenter (-9999 to +9999)

(2) Rectangular coordinates

The origin (+0000, +0000) is the left-top edge of a field of view.

(Example output)

10000000099070(+0269,+0011)(+0455,+0086)(+0386,+0270)(+0198,+0199)

(Output format)

Header + Data + (A1) + (A2) + (A3) + (A4) + Terminator
A1 - A4　　: (abbbb,acccc)
a　　: + or -
bbbb: X coordinates of four corners of a symbol (-9999 to +9999)
cccc: Y coordinates of four corners of a symbol (-9999 to +9999)

(3) Barycentric / Rectangular coordinates

Outputs coordinates of both barycenter and four corners of a symbol.

(Example output)

10000000099070(+0007,+0103)(+0269,+0011)(+0455,+0086)(+0386,+0270)(+0198,+0199)

(Output format)

Header + Data +(B) + (A1) + (A2) + (A3) + (A4) + Terminator

(B): Barycentric coordinate

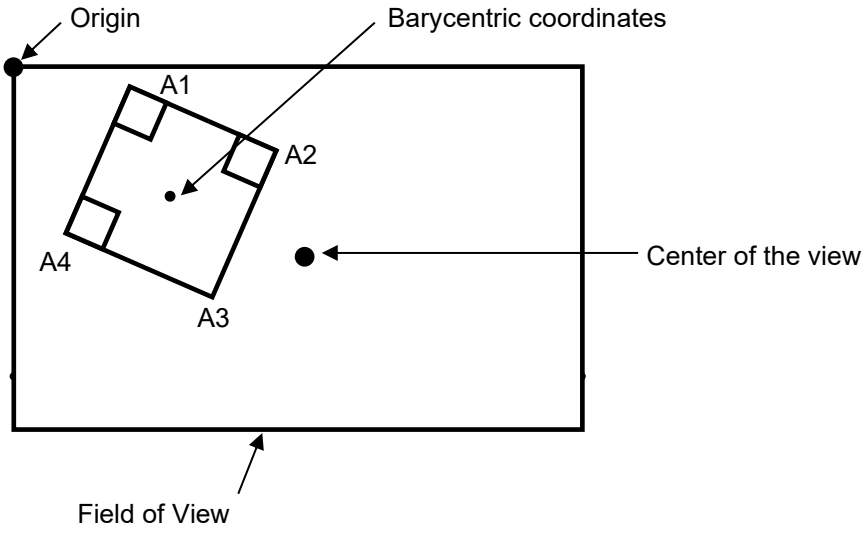
(A1)(A2)(A3)(A4): Rectangular coordinates

B, A1-A4: (abbbb,acccc)

a: + or -

bbbb: X coordinate of barycenter and X coordinate of four corners of a symbol (-9999 to +9999)

cccc: Y coordinate of barycenter and Y coordinate of four corners of a symbol (-9999 to +9999)



6.2.8 **Table Number**

Adds setting table number used in the last decoding.

(Example output)

10000000099070(TBL:3)

(Output format)

Header + Data + (TBL:a) + Terminator

a: Table number

6.2.9 **Reader Name**

Adds the individual name of the code reader.

(Example output)

(My Reader)10000000099070

(Output format)

Header + (a) + Data + Terminator

a: Reader Name

6.3 Save Image

Maximum 64 captured images can be stored. The number of cycle buffer memory means the number of storable images.

MAXIMG=a[CR] a=1 to 64 Specify number of cycle buffer memory.
MAXIMG=1[CR] (*)
*: Default

6.3.1 Transmit processed image

Select an image to save from the 2 images captured before or after preprocessing.
(Only for latest image)

IMGSEL=0[CR] Transmit captured image (Raw image)
IMGSEL=1[CR] (*) Transmit decoded image(*)
*: Default

6.3.2 Overwrite setting of cycle buffer memory

When the number of saved images exceeds the number of cycle buffer memories, you can select whether to overwrite the cycle buffer memory to continue imaging or to stop imaging without overwriting.

CAPLIM=0[CR] (*) Overwrites from the oldest image in the cycle buffer memory.
CAPLIM=1[CR] Interrupt the imaging.
*: Default

6.3.3 Preparation for transmitting the saved image

Using ”MCRConfig” software allows the download of the saved image that has been completed the preparation of transmission to the host. (only for USB type)

IMGFRAME=0,0 (*) Preparation for the last saved image
*: Default

Any number of the buffer can be set. The setting after each SYNC command is necessary because it is set to the default by a SYNC command.

IMGFRAME=0,a[CR] a=1 to 64
Preparation for the image saved in buffer No. (a)

6.3.4 **Output the settings of saving images**

```
?IMG[CR]              Output the status

 (Output example)
************* STATUS *************
 IMODE=a
 PX=b
 PY=c
 WX=d
 WY=e
 IMGSEL=f (0:captured image 1:decoded image)
 MAXIMG=g (1-64)
 WAITIMG=h (0-1000[ms])
 CAPLIM=i (0:OFF 1:ON)
 ROTATE=j (0:disable 1:enable)
 GOUTSIZE=k (0:752x480 1:376x240 2:188x120 3:94x60)
 IMGFRAME=0,l (0:fixed value, 1-64:frame-no.)
************** END ***************
MCR-F110U : 077-V1.1a

a             : Trimming
b,c,d,e       : Trimming position (Left, Top, Width, Height)
f             : Image type (Captured image, Decoded image)
g             : Number of cycle buffer memory
h             : Capturing time interval in the cycle buffer function
                (Unit: ms)
i             : Overwrite/not overwrite the cycle buffer memory when the
                number of captured image exceeds the number of cycle
                buffer memory
j             : Image rotation (0 degree/180 degrees)
k             : Image size for continuous image transfer
l             : Cycle buffer memory number
```

6.4 Save NG (No Read) Image

Up to 10 decoded images can be saved when reading is NG (No Read).
The NG Image is cleared when the power is turned off.
MCRConfig (version: 1.6.5 or later) supports the NG image acquisition function.

6.4.1 Set image type

This command specifies image type for the transfer.

IMGSEL=0[CR] Captured image (Raw image)
IMGSEL=1[CR] (*) Decoded image (Preprocessed image)

*: Default

6.4.2 NG Image storage order

Select the storage order when the number of NG Images exceeds 10.

IMGFULL=0[CR] Save the latest 10 NG Images

NG Image buffer No.									
1	2	3	4	5	6	7	8	9	10
1	2	3	4	5	6	7	8	9	10
11	12								

← storage order

IMGFULL=1[CR] (*) Save the 1st to 9th NG Images and the latest NG Image

NG Image buffer No.									
1	2	3	4	5	6	7	8	9	10
1	2	3	4	5	6	7	8	9	10
									11
									12

← storage Order

*: Default

6.4.3 **Checking the number of stored NG Images**

Outputs the number of saved NG Images (status).

?NGFRAME[CR]　　checking command

Sample Output

STS	Mark for starting output
NGNUM:6	Number of NG Images (6 NG Images can be extracted)
END	Mark of output completion

6.4.4 **Preparation for transmitting the NG Image**

Sets the NG image to be transferred to the host PC.

NGFRAME=a[CR]　　a=0 to 10

a=0 : Do not transfer NG Images (last decoded image) **(*)**
a=1-10 : Prepare to transfer the NG Image No. (a)
(the oldest image is No.1)

*: Default

6.5 PLC Link [Only for RS-232C type]

With PLC link, you can write a reading result of symbols directly to the PLC internal data memory (programmable logic controller) through RS-232C interface. For the details, download ”PLC Link Instruction Manual for Image Reader” from:

https://www.mars-tohken.co.jp/en/DL/

6.6 Simultaneous reading of multiple labels

In symbol reading mode, multiple (up to 10) labels can be read simultaneously.

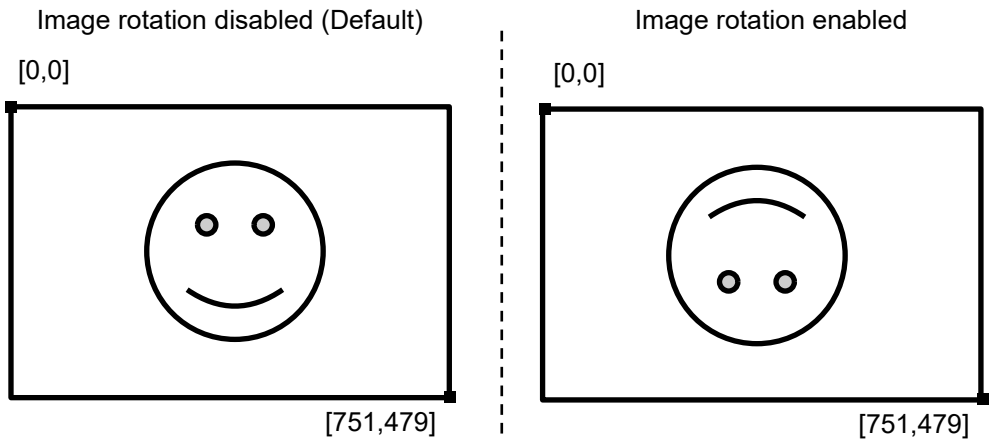
The order of reading data output can be configured like the following.

1. By completion of reading process
2. By preset number of digits
3. By preset matching characters
4. By preset symbologies

† Setting more symbols in an image than the number of preset symbols causes output failure.

6.7 Image Rotation

Rotates an image 180 degrees before a capture. The rotated image is used in all the process (decoding, saving data, etc.).



6.8 Register of SYNC ON/OFF command

The SYNC ON/OFF command of the code reader can be registered with any 8 alphanumeric characters (available characters are 0x20 to 0x7E in 7.18 Table of Character Code). If an entered arbitrary character string is the same as an existing serial command, the command is not registered.

- SYNC ON command setting

  SONCMD=a[CR] (a: Up to 8 alphanumeric characters)
  The character string a is registered as SYNC ON command.

  e.g.)
  SONCMD=FIND[CR] Register "FIND" as SYNC ON command.
  SONCMD=[CR] **(*)** Default value is not registered.
  *: Default

- SYNC OFF command setting

  SOFFCMD=a[CR] (a: Up to 8 alphanumeric characters)
  The character string a is registered as SYNC OFF command.

  e.g.)
  SOFFCMD=END[CR] Register "END" as SYNC OFF command.
  SOFFNCMD=[CR] **(*)** Default value is not registered.
  *: Default

Note: SYNC OFF command can only be used in Timeout Mode.

6.9 Save SETUP / Load SETUP

SETUP: Code reader configuration parameters

Up to 20 setups can be stored in the code reader. When the setup is loaded, the settings are reflected in the code reader.

6.9.1 Settings targeted by Setup Save / Read Function

Settings targeted by Setup function are as follows.

Term	Command	Description
Symbologies Common	GS1GS=	Convert characters which indicate the data delimiter of GS1 related symbols (FNC1)
	GS1AI=	Set the output method of GS1 AI
Symbologies Multi-Labels	LABELS=	Number of labels
	OUTFORM=	Setting of Output in the order
	SET=$Na-b	Set label number and verification digit number
	SET=$Ca-bbcc	Set label number and verification character
	SET=$Sa-b	Set label number and verification symbologies
Symbologies Code39	SET=AM	Set Disable/Enable Code39
	C39SS=	Start / Stop code output setting of Code39
	DIGIT=A	Set minimum and maximum digit numbers of Code39
	EDIT=A	Set data output range of Code39
	SYMHEAD=A	Set header character of Code39
	SYMFOOT=A	Set footer character of Code39
Symbologies Code128	SET=CM	Set Disable/Enable Code128
	DIGIT=C	Set minimum and maximum digit numbers of Code128
	EDIT=C	Set data output range of Code128
	SYMHEAD=C	Set header character of Code128
	SYMFOOT=C	Set footer character of Code128

Term	Command	Description
Symbologies Codabar	SET=FM	Set Disable/Enable Codabar
	CODASS=	Start / Stop code output setting of Codabar
	CODACS=	Start / Stop code output setting of Codabar
	CODACHK=	Checksum algorithm setting of Codabar
	DIGIT=F	Set minimum and maximum digit numbers of Codabar
	EDIT=F	Set data output range of Codabar
	SYMHEAD=F	Set header character of Codabar
	SYMFOOT=F	Set footer character of Codabar
Symbologies ITF	SET=IM	Set Disable/Enable ITF
	DIGIT=I	Set minimum and maximum digit numbers of ITF
	EDIT=I	Set data output range of ITF
	SYMHEAD=I	Set header character of ITF
	SYMFOOT=I	Set footer character of ITF
Symbologies Code93	SET=GM	Set Disable/Enable Code93
	DIGIT=G	Set minimum and maximum digit numbers of Code93
	EDIT=G	Set data output range of Code93
	SYMHEAD=G	Set header character of Code93
	SYMFOOT=G	Set footer character of Code93
Symbologies UPC/EAN/JAN	SET=EM	Set Disable/Enable UPC/EAN/JAN
	JE13SUM=	Checksum output setting of EAN/JAN (standard 13 digits)
	JE8SUM=	Checksum output setting of EAN/JAN (standard 8 digits)
	UASUM=	Checksum output setting of UPC-A
	UESUM=	UESUM=a Checksum output setting of UPC-E
	UPCANS=	Number system output setting of UPC-A
	UPCENS=	Number system output setting of UPC-E
	UPCE=	Checksum output setting of UPC-E
	UPCTX=	Output format setting of UPC-A
	DIGIT=E	Set minimum and maximum digit numbers of UPC/EAN/JAN
	EDIT=E	Set data output range of UPC/EAN/JAN
	SYMHEAD=E	Set header character of UPC/EAN/JAN
	SYMFOOT=E	Set footer character of UPC/EAN/JAN

Term	Command	Description
Symbologies GS1 Databar	SET=eM-RSS	Set Disable/Enable GS1 Databar
	DIGIT=e -RSS	Set minimum and maximum digit numbers of GS1 Databar
	EDIT=e -RSS	Set data output range of GS1 Databar
	SYMHEAD=e -RSS	Set header character of GS1 Databar
	SYMFOOT=e -RSS	Set footer character of GS1 Databar
Symbologies DataMatrix	SET=dM	Set Disable/Enable DataMatrix
	DIGIT=d	Set minimum and maximum digit numbers of DataMatrix
	EDIT=d	Set data output range of DataMatrix
	SYMHEAD=d	Set header character of DataMatrix
	SYMFOOT=d	Set footer character of DataMatrix
Symbologies QR Code	SET=QM	Set Disable/Enable QR Code
	DIGIT=Qb,c	Set minimum and maximum digit numbers of QR Code
	EDIT=Qb,c	Set data output range of QR Code
	SYMHEAD=Q	Set header character of QR Code
	SYMFOOT=Q	Set footer character of QR Code
Symbologies Composite	SET=eM-CC	Set Disable/Enable Composite
	DIGIT=e -CC	Set minimum and maximum digit numbers of Composite
	EDIT=e -CC	Set data output range of Composite
	SYMHEAD=e -CC	Set header character of Composite
	SYMFOOT=e -CC	Set footer character of Composite
Symbologies PDF417	SET=LM	Set Disable/Enable PDF417
	DIGIT=L	Set minimum and maximum digit numbers of PDF417
	EDIT=L	Set data output range of PDF417
	SYMHEAD=L	Set header character of PDF417
	SYMFOOT=L	Set footer character of PDF417
Symbologies Aztec Code	SET=zM	Set Disable/Enable Aztec Code
	DIGIT=z	Set minimum and maximum digit numbers of Aztec Code
	EDIT=z	Set data output range of Aztec Code
	SYMHEAD=z	Set header character of Aztec Code
	SYMFOOT=z	Set footer character of Aztec Code

Term	Command	Description
Reading	BR=	No read message
	SONCMD=	Register the SYNC ON command
	SOFFCMD=	Register the SYNC OFF command
Operation Mode	SYNCMODE	Set Operation Mode
	VOUT=	Output timing
	TOTALLIM=	Decode timeout
	continue/stop	Continuous Reading Mode Enable/Disable
	contintvl=	Set a time interval of reading
Additional Information	CHKBCR=	Warning Information
	SYMBOLTX=	Symbology Information
	DTTX=	Decoding time
	TTTX=	Total time of decoding
	AGCTX=	Swing Value
	TBLTX=	Table Number
	MYNAME=	Individual name
	NAMETX=	Individual name to output data
	CONTTX=	Contrast Information
	CSX=	X-coordinate point of calculation (Origin)
	CSY=	Y-coordinate point of calculation (Origin)
	CWX=	X-coordinate point of calculation (Width)
	CWY=	Y-coordinate point of calculation (Width)
	QTTX=	Quality Information
	MRTX=	Mirrored Image Information
	XYTX=	Symbol Position Information
	PNTX=	Last read Setup Number
Delay Time	DELAY=	Delay time between SYNC input and start capturing
Decode Area	DECMODE=	Set Decode Area
Cycle Buffer Function	MAXIMG=	Number of captured images
	WAITIMG=	Capture interval of time when MAXIMG is greater than 1
	CAPLIM=	After the number of captures exceeds the MAXIMG
Image rotation	ROTATE=	Enable/disable image rotating (180a) at capturing images
Camera Control Mode	AGC=	Use the camera setting

Term	Command	Description
Table Mode	ENABLEDDMTBL=	Specify Number of Tables
	DDMTBLNG=	Specify the operation of Table Mode
	DDMcapmode=	Specify capturing area of the table
	DDMlight=	Set ON/OFF of illumination output
	DDMflash=	Set the brightness of internal illumination source
	DDMshutt=	Set the shutter speed of the table
	DDMagain=	Set the Analog gain of the table
	DDMdgain=	Set the Digital gain of the table
	DDMblackval=	Setting for improving the contrast of a captured image
	DDMpreproc=	Set the type of preprocessing
	DDMmirror=	Set the mirrored image of the table
	DDMdecodelim=	Setting of decoding time limit
	DDMtblname=	This command specifies a name of the current table number
Auto Gain Control Mode	CAPMODE=	Specify capturing area
	LIGHT=	Set ON/OFF of illumination output
	FLASH=	Set the brightness of internal illumination source
	SHUTT=	Set the shutter speed
	AGAIN=	Set the Analog gain
	DGAIN=	Set the Digital gain
	BLACKVAL=	Setting for improving the contrast of a captured image
	IPFUNC0=	Set the type of preprocessing
	MIRROR=	Set the mirrored image
	DECODELIM=	Setting of decoding time limit
Preset	PREM=	Set the Preset Mode
	SET=PREN	Set the number of digits for comparing
	SET=PRED	Set the preset data

Term	Command	Description
Auto sense	LFILLUM=	Set pointer and illumination in detection state
	LFSENS=	Set detection sensitivity
	LFTOTALLIM=	Set read timeout time in Auto-sense Mode
	LFWAIT=	Set the wait time for the next detection
	VERIFY=	Set whether to permit continuous reading of the same symbol
	VWAIT=	Set the prohibited time when VERIFY=1
NG Image storage order	IMGFULL=	Set the storage order the NG Image

6.9.2 **Serial Command for Setup Save / Read Function**

The commands related to setup are as follows:

PSAVE=a[CR] (a: 1 to 20)
Saves the current settings of the code reader in the code reader as SETUP “a”.

PLOAD=a[CR] (a: 1 to 20)
Reflects the settings of SETUP in the code reader.
If there is no SETUP “a”, the setting change is not performed.

PERASE[CR] Erase the setting of all SETUPs in the code reader. †

PERASE=a[CR] (a: 1 to 20) †
Erase the setting of SETUP “a” in the code reader.

PNAME=a,b[CR] (a: 1 to 20, b: Alphanumeric character 32 bytes)
Sets a name to SETUP “a”.

WSETP[CR] Saves the SETUP name in the core reader.

PNTX=a[CR] (a: 0 to 1) †
After power up, the read number of the SETUP is added to the read data.
When the setup is not read, "--" is added and added.
The output example is as follows.

12345(SETUP:01)

?plist[CR] Outputs the SETUP name.

(Output example of ?plist)

```
?plist
PNAME=1,SETUP1
PNAME=2,SETUP2
----------------------------(omission)-----------------------------
PNAME=20,
```

† This function is supported from firmware version V1.2a or later.

7 Serial Commands (RS-232C, USB)

- The following serial commands can be used through both RS-232C and USB interface unless otherwise specified.

  (Serial command format)

  Serial command + [CR]　　(*): Default (Backward compatible)

  [Header] + Serial command + [Terminator]

- To add header and terminator characters to a serial command, send the command "CMDFORM=1". The header and terminator characters are same as the ones to add to decoded data. If there is no terminator, it is compatible with the conventional operation.

- To return the input format of the configuration commands to compatible (configuration command + [CR]), send the command ”CMDFORM=0”.

7.1 Communication settings

Term	Command	Description (*): Default
Baud rate	BAUD=a	Baud rate of RS-232C a=1200: 1200bps, a=2400: 2400bps, a=4800: 4800bps a=9600: 9600bps (*), a=19200: 19200bps, a=38400: 38400bps a=57600: 57600bps, a=115200: 115200bps
Frame		(Length, Parity, Stop bit)
	FRAME=0	7 bits, Odd, 1 bit
	FRAME=1	7 bits, Odd, 2 bits
	FRAME=2	7 bits, Even, 1 bit
	FRAME=3	7 bits, Even, 2 bits
	FRAME=4	8 bits, None, 1 bit (*)
	FRAME=5	8 bits, None, 2 bits
	FRAME=6	8 bits, Parity, 1 bit
	FRAME=7	8 bits, Parity, 2 bits
	FRAME=8	8 bits, Even, 1 bit
	FRAME=9	8 bits, Even, 2 bits
Header	HEAD=a	Header character a=0; None (*) a=1; [SX] a=2; [EC]
Terminator	TERM=a	Terminator character a=0: None a=1: [EX] a=2: [EX][CR] a=3: [CR][LF] (*) a=4: [CR] a=5: [CR][TAB] a=6: [TAB][CR] a=7: [TAB]
Command Format	CMDFORM=0 CMDFORM=1	Command+[CR] or Command+[CR][LF] (*) [Header]+Command+[Terminator]
Separator	SEPA=a	Character code a=0: None a=1: '&' a=2: ,(Comma) (*) a=3: [FS] (1C hex) a=4: [GS] (1D hex) a=5: [SP] (20 hex)
Prefix	PREFIX =a =a,¥bb	Prefix character (Identifier) a=0: Disable (*), a=1: Enable bb: Identifier (Hex code (*1)) bb=00 (*) [NUL] When omitting " bb", configured identifier will not be changed.
Suffix	SUFFIX =a =a,¥bb	Suffix character (Identifier) a=0: Disable (*), a=1: Enable bb: Suffix character (Hex code (*1)) bb=00 (*) [NUL] When omitting " bb", configured identifier will not be changed.
RS/CS control	RSCS=0 RSCS=1	Disable RTS/CTS flow control (*) Enable RTS/CTS flow control
Control code	LABELTX=0 LABELTX=1	Output control code as it is (*) Output control code in hexadecimal[Hex]
Timeout period	TXWAIT=a	Timeout period for data transmission a=1000-30000 [ms], 10ms step 1000[ms] (*)
Command response	CMDRES=a	Command response a=0: Disable, a=1: Enable (*)

(*1) Refer to 7.18 "Table of Character Code".

7.2 Symbologies settings

Term	Command	Description (*): Default
Common	SET=#M0 SET=#M1	Disable all symbologies reading Enable all symbologies reading
	SYMHEAD =#a =#a,¥bb	Header character (Identifier) for all symbologies a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) When omitting " bb", configured identifier will not be changed.
	SYMFOOT =#a =#a,¥bb	Footer character (Identifier) for all symbologies a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) When omitting " bb", configured identifier will not be changed.
	GS1GS =a =a,¥bb	Convert characters (FNC1) that indicates the data delimiter (Identifier) of GS1 symbols a=0: Disable a=1: Enable (*) bb: Delimiter (Hex code (*1)) bb=1D (*) [GS] When omitting " bb", configured identifier will not be changed.
	GS1AI=a	Put the application identifier of GS1 symbol in brackets a=0: Disable (*) a=1: Enable
Multiple labels	LABELS=a	Number of labels a=1-10 a=1 (*)
	OUTFORM=0 OUTFORM=1 OUTFORM=2 OUTFORM=3	Output data in order of decoding (*) Output data in specified digit order Output data in specified character order Output data in order of specified symbologies
	SET=$Na-b	Set label number and verification digit number a=1-10: Label number, b=0-9999: Digit number
	SET=$Ca-bbcc	Set label number and verification character a=1-10: Label number, bb: The first character in Hex code (*1) cc: The second character in Hex code (*1)
	SET=$Sa-b	Set label number and verification symbologies a=1-10: Labe number, b: Symbol identifier (as follows) A:Code39 C:Code128 F:Codabar I:ITF G:Code93 E:JAN/EAN/UPC R:GS1 Databar c:Composite Code d:DataMatrix Q:QR Code L:PDF417 z:Aztec Code

(*1) Refer to 7.18 "Table of Character Code".

Term	Command	Description (*): Default
Code39	SET=AM0	Disable Code39
	SET=AM1	Enable Code39; disable C/D check (*)
	SET=AM2	Enable Code39; enable C/D check and output
	SET=AM3	Enable Code39; enable C/D check; disable C/D output
	SET=AM4	Enable Code39; disable C/D check; enable FULL ASCII mode
	SET=AM5	Enable Code39; enable C/D check and output; enable FULL ASCII mode
	SET=AM6	Enable Code39; enable C/D check; disable C/D output; enable FULL ASCII mode
	C39SS=0	Disable Code39 start/stop code output (*)
	C39SS=1	Enable Code39 start/stop code output
	DIGIT=Ab,c	Set the minimum number of digits (b) and maximum number of digits (c) of Code39
	EDIT=Ab,c	Set the start digit (b) and number of digits (c) of Code39 decoded data output
	SYMHEAD	Header character of Code39
	=Aa	a=0: Disable (*) a=1: Enable
	=Aa,¥bb	bb: Identifier (Hex code (*1)) bb=41 (*) **character "A"**
		When omitting " bb", configured identifier will not be changed.
	SYMFOOT	Footer character of Code39
	=Aa	a=0: Disable (*) a=1: Enable
	=Aa,¥bb	bb: Identifier (Hex code (*1)) bb=41 (*) **character "A"**
		When omitting " bb", configured identifier will not be changed.
Code128 GS1-128	SET=CM0	Disable Code128
	SET=CM1	Enable Code128 (*)
	DIGIT=Cb,c	Set the minimum number of digits (b) and maximum number of digits (c) of Code128
	EDIT=Cb,c	Set the start digit (b) and number of digits (c) of Code128 decoded data output
	SYMHEAD	Header character of Code128
	=Ca	a=0: Disable (*) a=1: Enable
	=Ca,¥bb	bb: Identifier (Hex code (*1)) bb=43 (*) **character "C"**
		When omitting " bb", configured identifier will not be changed.
	SYMFOOT	Footer character of Code128
	=Ca	a=0: Disable (*) a=1: Enable
	=Ca,¥bb	bb: Identifier (Hex code (*1)) bb=43 (*) **character "C"**
		When omitting " bb", configured identifier will not be changed.
Codabar	SET=FM0	Disable Codabar
	SET=FM1	Enable Codabar; disable C/D check (*)
	SET=FM2	Enable Codabar; enable C/D check and output
	SET=FM3	Enable Codabar; enable C/D check; disable C/D output
	CODASS=0	Disable Codabar start/stop code output
	CODASS=1	Enable Codabar start/stop code output (*)
	CODACS=0	Set Codabar start/stop code to lower case
	CODACS=1	Set Codabar start/stop code to upper case (*)
	CODACHK=0	Codabar check digit calculation method: modulus16 (*)
	CODACHK=1	Codabar check digit calculation method: 7DR
	DIGIT=Fb,c	Set the minimum number of digits (b) and maximum number of digits (c) of Codabar

Codabar (continued)	EDIT=Fb,c	Set the start digit (b) and number of digits (c) of Codabar decoded data output
	SYMHEAD =Fa =Fa,¥bb	Header character of Codabar a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=46 (*) character "F" When omitting " bb", configured identifier will not be changed.
	SYMFOOT =Fa =Fa,¥bb	Footer character of Codabar a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=46 (*) character "F" When omitting " bb", configured identifier will not be changed.
ITF	SET=IM0 SET=IM1 SET=IM2 SET=IM3	Disable ITF Enable ITF; disable C/D check (*) Enable ITF; enable C/D check and output Enable ITF; enable C/D check; disable C/D output
	DIGIT=Ib,c	Set the minimum number of digits (b) and maximum number of digits (c) of ITF
	EDIT=Ib,c	Set the start digit (b) and number of digits (c) of ITF decoded data output
	SYMHEAD =Ia =Ia,¥bb	Header character of ITF a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=49 (*) character "I" When omitting " bb", configured identifier will not be changed.
	SYMFOOT =Ia =Ia,¥bb	Footer character of ITF a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=49 (*) character "I" When omitting " bb", configured identifier will not be changed.
Code93	SET=GM0 SET=GM1	Disable Code93 Enable Code93 (*)
	DIGIT=Gb,c	Set the minimum number of digits (b) and maximum number of digits (c) of Code93
	EDIT=Gb,c	Set the start digit (b) and number of digits (c) of Code93 decoded data output
	SYMHEAD =Ga =Ga,¥bb	Header character of Code93 a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=47 (*) character "G" When omitting " bb", configured identifier will not be changed.
	SYMFOOT =Ga =Ga,¥bb	Footer character of Code93 a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=47 (*) character "G" When omitting " bb", configured identifier will not be changed.
UPC/EAN/JAN	SET=EM0 SET=EM1	Disable UPC/EAN/JAN Enable UPC/EAN/JAN (*)
	JE13SUM=0 JE13SUM=1	EAN/JAN (standard 13 digits); enable checksum output(*) EAN/JAN (standard 13 digits); disable checksum output
	JE8SUM=0 JE8SUM=1	EAN/JAN (standard 8 digits); enable checksum output(*) EAN/JAN (standard 8 digits); disable checksum output
	UASUM=0 UASUM=1	UPC-A; enable checksum output (*) UPC-A; disable checksum output
	UESUM=0 UESUM=1	UPC-E; enable checksum output (*) UPC-E; disable checksum output

UPC/EAN/JAN (continued)	UPCANS=0 UPCANS=1	UPC-A; enable number system character output (*) UPC-A; disable number system character output
	UPCENS=0 UPCENS=1	UPC-E; enable number system character output (*) UPC-E; disable number system character output
	UPCE=0 UPCE=1	UPC-E; output in UPC-A format (*) UPC-E; output in UPC-E format
	UPCTX=0 UPCTX=1	UPC-A; disable "0" addition at the beginning of data (*) UPC-A; enable "0" addition at the beginning of data
	DIGIT=Eb,c	Set the minimum number of digits (b) and maximum number of digits (c) of UPC/EAN/JAN
	EDIT=Eb,c	Set the start digit (b) and number of digits (c) of UPC/EAN/JAN decoded data output
	SYMHEAD =Ea =Ea,¥bb	Header character of UPC/EAN/JAN a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=45 (*) character "E" When omitting " bb", configured identifier will not be changed.
	SYMFOOT =Ea =Ea,¥bb	Footer character of UPC/EAN/JAN a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=45 (*) character "E" When omitting " bb", configured identifier will not be changed.
GS1 DataBar	SET=eM00-RSS SET=eM01-RSS SET=eM02-RSS SET=eM04-RSS SET=eM08-RSS SET=eM10-RSS SET=eM1F-RSS	Disable GS1 DataBar (*) Enable GS1 DataBar; Expanded Enable GS1 DataBar; Expanded Stacked Enable GS1 DataBar; Limited Enable GS1 DataBar; Omni directional, Truncated Enable GS1 DataBar; Stacked, Stacked Omni directional Enable GS1 DataBar; All types
	DIGIT=eb,c-RSS	Set the minimum number of digits (b) and maximum number of digits (c) of GS1 DataBar
	EDIT=eb,c-RSS	Set the start digit (b) and number of digits (c) of GS1 DataBar decoded data output
	SYMHEAD =ea,¥bb-RSS	Header character of GS1 DataBar a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=52 (*) character "R" When omitting " bb", configured identifier will not be changed.
	SYMFOOT =ea,¥bb-RSS	Footer character of GS1 DataBar a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=52 (*) character "R" When omitting " bb", configured identifier will not be changed.

(*1) Refer to 7.18 "Table of Character Code".

Term	Command	Description (*): Default
DataMatrix	SET=dM0	Disable DataMatrix
	SET=dM1	Enable DataMatrix; Square, Normal label
	SET=dM2	Enable DataMatrix; Square, Black/White inverted label
	SET=dM3	Enable DataMatrix; Square, B/W inverted and Normal label
	SET=dM5	Enable DataMatrix; Square and Rectangle, Normal label
	SET=dM6	Enable DataMatrix; Square and Rectangle, B/W inverted and Normal label
	SET=dM7	Enable DataMatrix; All types (*)
	DIGIT=db,c	Set the minimum number of digits (b) and maximum number of digits (c) of DataMatrix
	EDIT=db,c	Set the start digit (b) and number of digits (c) of DataMatrix decoded data output
	SYMHEAD =da =da,¥bb	Header character of DataMatrix a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=64 (*) character "d" When omitting " bb", configured identifier will not be changed.
	SYMFOOT =da =da,¥bb	Footer character of DataMatrix a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=64 (*) character "d" When omitting " bb", configured identifier will not be changed.
QR Code Micro QR Code	SET=QM0	Disable QR Code
	SET=QM1	Enable QR Code; Normal label
	SET=QM2	Enable QR Code; Black/White inverted label
	SET=QM3	Enable QR Code: B/W inverted and Normal label (*)
	DIGIT=Qb,c	Set the minimum number of digits (b) and maximum number of digits (c) of QR Code
	EDIT=Qb,c	Set the start digit (b) and number of digits (c) of QR Code decoded data output
	SYMHEAD =Qa =Qa,¥bb	Header character of QR Code a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=51 (*) character "Q" When omitting " bb", configured identifier will not be changed.
	SYMFOOT =Qa =Qa,¥bb	Footer character of QR Code a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=51 (*) character "Q" When omitting " bb", configured identifier will not be changed.
Composite Code	SET=eM00-CC	Disable Composite (*)
	SET=eM01-CC	Enable EAN/UPC Composite
	SET=eM02-CC	Enable GS1 DataBar Composite
	SET=eM04-CC	Enable GS1-128 Composite
	SET=eM07-CC	Enable all types of Composite
	DIGIT=eb,c-CC	Set the minimum number of digits (b) and maximum number of digits (c) of Composite
	EDIT=eb,c-CC	Set the start digit (b) and number of digits (c) of Composite decoded data output
	SYMHEAD =ea,¥bb-CC	Header character of Composite a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=63 (*) character "c" When omitting " bb", configured identifier will not be changed.

Composite Code (continued)	SYMFOOT =ea,¥bb-CC	Footer character of Composite a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=63 (*) character “c” When omitting ” bb”, configured identifier will not be changed.
PDF417	SET=LM0 SET=LM1	Disable PDF417 Enable PDF417 (*)
	DIGIT=Lb,c	Set the minimum number of digits (b) and maximum number of digits (c) of PDF417
	EDIT=Lb,c	Set the start digit (b) and number of digits (c) of PDF417 decoded data output
	SYMHEAD =La =La,¥bb	Header character of PDF417 a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=64 (*) character “L” When omitting ” bb”, configured identifier will not be changed.
	SYMFOOT =La =La,¥bb	Footer character of PDF417 a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=64 (*) character “L” When omitting ” bb”, configured identifier will not be changed.
Aztec Code	SET=zM0 SET=zM1 SET=zM2 SET=zM3	Disable Aztec Code (*) Enable Aztec Code; Normal label Enable Aztec Code; Black/White inverted label Enable Aztec Code; B/W inverted and Normal label
	DIGIT=zb,c	Set the minimum number of digits (b) and maximum number of digits (c) of Aztec Code
	EDIT=zb,c	Set the start digit (b) and number of digits (c) of Aztec Code decoded data output
	SYMHEAD =za =za,¥bb	Header character of Aztec Code a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=51 (*) character “z” When omitting ” bb”, configured identifier will not be changed.
	SYMFOOT =za =za,¥bb	Footer character of Aztec Code a=0: Disable (*) a=1: Enable bb: Identifier (Hex code (*1)) bb=51 (*) character “z” When omitting ” bb”, configured identifier will not be changed.

(*1) Refer to 7.18 “Table of Character Code”.

7.3 Symbol Reading (Operation, Adjustment and Diagnostic mode)

Term	Command	Description (*): Default
Reading	g	Start reading (SYNC ON)
	BR=a	Read error message a=0: None a=1: "BR" (*) a=2: "ERROR" a=3: "?" a=4: None (Header, Terminator and Additional information are output) a=5: "NO READ"
	SONCMD=a	Register the SYNC ON command (alphanumeric 8 digits) a=None (*)
	SOFFCMD=a	Register the SYNC OFF command (alphanumeric 8 digits) a=None (*)
Operation Mode	SYNCMODE=a	a=0: Normal Mode (*) a=1: Reading Timeout Mode a=2: External Trigger Mode a=3: Auto-sense Mode
	VOUT=a	Data output timing a=0: After decoding (*) a=1: After SYNC off
	TOTALLIM=a	Decode timeout a=100-30000 [ms], 10[ms] step a=3000[ms](*)
Continuous Reading Mode	continue	Enable Continuous Reading Mode
	stop	Exit Continuous Reading Mode (back to Normal Mode) (*)
	contintvl=a	Set the interval to the next reading a=0-10000 [ms] a=1000(*)
Test Mode	TEST=a	a=0: Normal Mode (*) a=1: Test Mode
	TMN=a	Number of reading times a=10-1000 (10 steps) a=10(*)
I/O Test	IOTEST=a,b	Set output time a=D1: OUT1 port D2: OUT2 port b=0: OFF(*) 10-10000: ON time, Units: ms -1: Remain ON
Reading Statistics	?total	Check the number of SYNC, Read OK and Read NG since power-on
	clrtotal	Clear memory
Decoding time	?dect	Check the decode time (minimum, maximum, average) of Read OK and Read NG from power-on
	clrdect	Clear memory
Warning Information	CHKBCR=a	a=0: Disable addition of information to read data (*) a=1: Enable addition of information to read data
Symbology Information	SYMBOLTX=a	a=0: Disable addition of information to read data (*) a=1: Enable addition of information to read data
Decoding time	DTTX=a	a=0: Disable addition of information (*) a=1: Enable addition of information
Total time of decoding	TTTX=a	a=0: Disable addition of information (*) a=1: Enable addition of information Info.) Number of decoded images and decoding time
Shutter speed, Analog gain and Digital gain	AGCTX=a	a=0: Disable addition of information (*) a=1: Enable addition of information

Table Number	TBLTX=a	a=0: Disable addition of information (*) a=1: Enable addition of information
Reader Name	MYNAME=a	Set reader name (31 characters or less) (*) My Reader
Addition of reader name	NAMETX=a	a=0: Disable addition of information (*) a=1: Enable addition of information
Contrast Information	CONTTX=a	a=0: Disable addition of information (*) a=1: Enable addition of information
	CSX=a	X coordinate of base point of calculation area for Read NG a=0-742 a=0(*)
	CSY=a	Y coordinate of base point of calculation area for Read NG a=0-480 a=0(*)
	CWX=a	X coordinate of width of calculation area for Read NG a=10-752 a=752(*)
	CWY=a	Y coordinate of width of calculation area for Read NG a=10-480 a=480(*)
Quality Information	QTTX=a	a=0: Disable addition of information (*) a=1: Enable addition of information
Mirrored Image Information	MRTX=a	a=0: Disable addition of information (*) a=1: Enable addition of information Add "MR-ON" when it is a mirrored image Add "MR-OFF" when it is not a mirrored image
Symbol Position Information	XYTX=a	a=0: Disable addition of information (*) a=1: Enable addition of barycentric coordinate data a=2: Enable addition of rectangular coordinate data a=2: Enable addition of barycentric and rectangular coordinate data
Setup Number Information	PNTX=a	a=0: Disable addition of information (*) a=1: Enable addition of information
Delay time	DELAY=a	Delay time from SYNC ON to exposure start a=0-30000 [ms] a=0 (*)
Decode Area	DECMODE= 0, Left, Top, Width, Height	Set decode area Not link with captured area Left 0-742 0 (*) Top 0-470 0 (*) Width 10-752 752 (*) Height 10-480 480 (*)
Cycle Buffer Function	MAXIMG=a	Number of captured images a=1-64 a=1(*)
	WAITIMG=a	Capture interval time a=0-1000 [ms] a=100 (*)
	CAPLIM=a	Operation setting when the number of captures exceeds the MAXIMG a=0: Overwrites cycle buffer memory and continue capturing (*) a=1: Stop capturing without overwriting the cycle buffer memory

7.4 Auto-Tuning

Term	Command	Description (*): Default
Auto-Tuning	*setup	Start Auto-Tuning
Cancel	*cancel	Cancel running Auto-Tuning
Auto-Tuning Status	*?state	Get the status of running Auto-Tuning STANDBY Standby RECEIVED Running Auto-Tuning SAVEWAIT Waiting for saving the tuning result ABORTED Auto-Tuning failed (No reply) Reading
Save Tuning Result (3 Types)	*save	Save the tuning result to Table #1 to #3
Save Tuning Result (1 type)	*save2=a	Save the tuning result to the specified table a=1-16

7.5 Camera Control (1)

Term	Command	Description (*): Default
Image Rotation	ROTATE=a	Rotation of captured image a=0: Disable rotation (*) a=1: Enable rotation

7.6 Camera Control (2) (for Table Mode)

Term	Command	Description (*): Default
Camera Control Mode	AGC=T	Use camera setting in Table Mode (*)
Number of Tables	ENABLEDDMTBL=a	Number of tables to enable a=1-16 a=16 (*) In case of a=3, Table #1, #2 and #3 are enabled and Table #4 through #16 are disabled. In case of a=1, Table #1 is enabled. The shutter speed, digital gain and the illumination setting are fixed.
Number of edit table	EDITDDMTBL=a	Specify the table to be edited a=1-16 a=1 (*) e.g.) When setting DDMagain value of Table #3 to 2, enter the command as follows. EDITDDMTBL=3 DDMagain=2
Copy Table settings	CPDDMTBL=a,b or a,b-c	Copy the settings of Table #a to Table #b Copy the settings of Table #a from Table #b to Table #c
Operation of Table Mode	DDMTBLNG=a	Specify the operation of Table Mode a=0: Start reading from the table number that decoded last. If the decode failed, start reading from the next table number. a=1: Start with Table #1 every time (*)

7.7 Camera Control (3) (for Table Mode)

Term	Command	Description (*): Default
Capturing area	DDMcapmode= Left, Top, Width, Height	Specify capturing area of the Table Left 0-742 0(*) Top 0-470 0(*) Width 10-752 752(*) Height 10-480 480(*)
Illumination Control	DDMlight=a	Specify illumination ON/OFF setting of the Table a=0: OFF a=3: ON (When imaging) (*) a=10: ON (Always) a=11: ON (During attempting to read: SYNCMODE=1 or 2)
Illumination brightness	DDMflash=a,b	Specify the illumination brightness a=0 (OFF), 1 (dark) ～ 100 (bright) a=100 (*) b=0 (OFF), 1 (dark) ～ 100 (bright) b=100 (*) When a value in the range of 1 to 100 is set, set a=b. When b is omitted, b=a is set.
Shutter speed	DDMshutt=a	Specify the shutter speed of the Table a=0-9 0 : 1/60 [sec] 1 : 1/125 2 : 1/250 (*) 3 : 1/500 4 : 1/1000 5 : 1/2000 6 : 1/4000 7 : 1/6000 8 : 1/8000 9 : 1/10000
Analog Gain	DDMagain=a	Specify the analog gain of the Table a=1 (Lowest) to 25 (Highest) a=9 (*)
Digital Gain	DDMdgain=a	Specify the digital gain of the Table a=1 (Lowest) to 12 (Highest) a=1 (*)
Black Level adjustment	DDMblackval=a	For improving the contrast of a captured image a=1 (Black emphasis) to 255 (White emphasis) a=128 (*)
Image Preprocessing	DDMpreproc= a,b,c,d,e	Set the type of preprocessing a,b,c,d,e : Command for preprocessing (Described later) a,b,c,d,e =0,0,0,0,0 (*) Up to 5 stage of function can be set
Mirrored Image	DDMmirror=a	Set reading of mirrored image of the Table a=0: Read only normal type (not mirrored) a=1: Read only mirrored type a=2: Read normal type and if it fails, try to read it as mirrored type (*)
Decoding time limit	DDMdecodelim=a	Set time limit for decoding process in the Table a=50-10000 [ms], 10[ms] step a=500 (*)
Table Name	DDMtblname=a	Set table name a=Table name (16 characters or less) (*)name

Note: The default values are the same in Table #1 to #16

7.8 Camera Control (3) (for Auto Gain Control Mode)

Term	Command	Description (*): Default
Camera Control Mode	AGC=E	Use camera setting in Auto Gain Control Mode
Capture area	CAPMODE= Left, Top, Width, Height	Specify capturing area Left 0-742 0 (*) Top 0-470 0 (*) Width 10-752 752 (*) Height 10-480 480 (*)
Illumination Control	LIGHT=a	Specify illumination ON/OFF setting a=0: OFF a=3: ON (When imaging) (*) a=10: ON (Always) a=11: ON (During attempting to read: SYNCMODE=1 or 2)
Illumination brightness	FLASH=a,b	Specify the illumination brightness a=0 (OFF), 1 (dark) ～ 100 (bright) a=100 (*) b=0 (OFF), 1 (dark) ～ 100 (bright) b=100 (*) When a value in the range of 1 to 100 is set, set a=b. When b is omitted, b=a is set. Specify the illumination brightness
Shutter speed	SHUTT=a	Specify the shutter speed a=0-9 0 : 1/60 [sec] 1 : 1/125 2 : 1/250 (*) 3 : 1/500 4 : 1/1000 5 : 1/2000 6 : 1/4000 7 : 1/6000 8 : 1/8000 9 : 1/10000
Analog Gain	AGAIN=a	Specify the analog gain a=1 (Lowest) to 25 (Highest) a=9 (*)
Digital Gain	DGAIN=a	Specify the digital gain a=1 (Lowest) to 12 (Highest) a=1 (*)
Black Level adjustment	BLACKVAL=a	For improving the contrast of a captured image a=1 (Black emphasis) to 255 (White emphasis) a=128 (*)
Image Preprocessing	IPFUNCO= a,b,c,d,e	Set the type of preprocessing a,b,c,d,e: Command for preprocessing (Described later) a,b,c,d,e=0,0,0,0,0 (*) Up to 5 stage of function can be set
Mirrored Image	MIRROR=a	Set reading of mirrored image of the Table a=0; Read only normal type (not mirrored) a=1; Read only mirrored type a=2; Read normal type and if it fails, try to read it as mirrored type (*)
Decoding time limit	DECODELIM=a	Set time limit for decoding process a=50-10000 [ms], 10[ms] step a=500 (*)

7.9 Image Preprocessing

Term	Command	Description
None	0	No Image Preprocessing
Black Erosion	1	Erosion 2x2
	3	Erosion 3x3
	5	Erosion 5x5
	7	Erosion 2x1 (w)
	9	Erosion 1x2 (h)
White Erosion	2	Dilation 2x2
	4	Dilation 3x3
	6	Dilation 5x5
	8	Dilation 2x1 (w)
	10	Dilation 1x2 (h)
Reverse	20	White & Black Reverse
Contrast Enhancement	21	Cont(L12cut): Truncate the 12% of dark side
	22	Cont(L22cut): Truncate the 25% of dark side
	23	Cont(H23cut): Round up the 12% of bright side
	24	Cont(H24cut): Round up the 25% of bright side
	25	Cont(LH12cut): Truncate the 12% of dark side and round up the 12% of bright side
	26	Cont(LH22cut): Truncate the 25% of dark side and round up the 25% of bright side
Gamma Correction	27	Gamma(r=2.0): Perform a Gamma Correction (2.0)
Smoothing Filter	40	Low pass 3x3 Smoothing Filter, Average value
Median Filter	41	Median filter 3x3, Median value
Down Size	70	Down Size (Quick): Image reduction (speed priority)
Down Size	71	Down Size (Average): Image reduction (luminance average)

7.10 Preset Mode

Term	Command	Description (*): Default
Preset	?pre	Get the status of Preset Mode settings
	PREM=0	Preset Mode 0: Disable Preset Mode (*)
	PREM=1	Preset Mode 1: Compare with the first read data after power on
	PREM=2	Preset Mode 2: Compare with registered data
	SET=PRENa	Set the number of digits for comparing a=0 to 100, a=0: No count
	clrpren	Clear the value of SET=PRENa
	SET=PREDa	Set the preset data
	clrpred	Clear the value of SET=PREDa

7.11 PLC Link

Term	Command	Description (*): Default
PLC Link Start/Exit	PLCMODE=a	Start/Exit PLC Link function a=0: Exit PLC Link Mode (*) a=2: Mitsubishi MELSEC Q/L series 232C MC protocol QnA compatible 3C frame (ASCII code) a=4; OMRON SYSMAC series 232C C mode command a=6: Mitsubishi MELSEC FX series 232C MC protocol A compatible 1C frame (ASCII code) a=8: KEYENCE KV series 232C KV command mode a=10; Panasonic FP7 series 232C MEWTOCOL-COM
Leading address	PLCDTA=a	Set the leading address a=0 to 32767 a=0 (*)
Padding character	PLCPAD=a	Set the padding character to be added when the number of digits in the read data is odd a=0: NULL [0x00] (*) a=1: SPACE [0x20]
SYNC IN by trigger area	PLCTRIG=a	Set SYNC IN by PLC using trigger area a=0: Enable the SYNC IN by trigger area (*) a=1: Disable the SYNC IN by trigger area
Trigger area monitoring interval	PLCINTVL=a	Set the interval to the next monitoring a=10-500 [ms] a=10 (*)
Trigger area monitoring start wait time	PLCTIM=a	Set the time to start monitoring after setting PLCTRIG=0 a=5-20 [s] a=10 (*)

7.12 Auto sense

Term	Command	Description (*): Default
Pointer / Illumination	LFILLUM=a	Set pointer and illumination in detection state a=0: Pointe OFF, Illumination OFF a=1; Pointer ON, Illumination OFF (*) a=2; Pointer OFF, Illumination ON
Detection sensitivity	LFSENS=a	Set detection sensitivity a=0: High a=1: Slightly high a=2: Normal (*) a=3: Slightly low a=4: Low
Read timeout time	LFTOTALLIM=a	Set read timeout time in Auto-sense Mode a=1-10 [s] a=3 (*)
Wait time	LFWAIT=a	Set the wait time for the next detection a=1-10 [s] a=1 (*)
Continuous reading of the same symbol	VERIFY=a	Set whether to permit continuous reading of the same symbol. a=0: Permission (*) a=1: Prohibition
Wait time for continuous reading of the same symbol	VWAIT=a	Set the prohibited time when VERIFY=1. a=1-20, Unit: x 500[ms] a=3 (*)

7.13 Image Output

Term	Command	Description (*): Default
Output target of image	Mode=a	Set the image transfer after reading a=0 : Disable (*) a=1 or 2: Reserved. DO NOT USE a=3 : Output OK/NG image in binary format a=4 : Output NG image in binary format a=5 : Output OK image in binary format
Output image size	GOUTSIZE=a	Set the image size a=0: 752x480 (*) a=1: 376x240 a=2: 188x120 a=3: 94x60
Image type	IMGSEL=a	Select image type a=0: Captured image (raw data) a=1: Decode image (processing data) (*)
Trimming	IMODE=a	a=0: Disable trimming (*) a=1: Enable trimming
Trimming position	 PX=a PY=a WX=a WY=a	Set the area of image to trim Start position (X coordinate) a=0-751 a=0 (*) Start position (Y coordinate) a=0-479 a=0 (*) Width (X direction) a=1-752 a=752 (*) Width (Y direction) a=1-480 a=480 (*)

7.14 Save NG (No Read) Image

Term	Command	Description (*): Default
Image type	IMGSEL=a	Select image type a=0: Captured image (raw data) a=1: Decode image (processing data) (*)
Storage order	IMGFULL=a	a=0: Save the latest 10 NG Images a=1: Save the 1st to 9th NG Image and the latest NG Image (*)
Number of stored NG Images	?NGFRAME	Checking the number of stored NG Images
Preparation for transmitting	NGFRAME=a	a=0: Do not transfer NG Images (last decoded image) (*) a=1-10: Prepare for transfer of specified NG image

7.15 Configuration Reference

Term	Command	Description
Configuration reference	?	Status transmission (First page)
	?2 or ??	Status transmission (Second page)
	?3 or ???	Status transmission (Third page)
	?4	Status transmission (Fourth page)
	?5	Status transmission (Fifth page)
	?img or ?IMG	Status transmission (Settings for image output)
	?ddmtbl or ?DDMTBL	Status transmission (Settings for Table mode)
	?plc or ?PLC	Status transmission (Settings for PLC Link mode)
	?lf or ?LF	Status transmission (Settings for Auto-sense mode)
	?pre or ?PRE	Status transmission (Settings for Preset mode)
	?tx or ?TX	Status transmission (Settings for additional information)

7.16 Setup

Term	Command	Description
Setup	PSAVE=a	Save current settings as SETUP-a in flash memory a=1-20
	PLOAD=a	Load SETUP-a settings saved in flash memory a=1-20
	PERASE	Erase the setting of all SETUPs in the code reader.
	PERASE=a	Erase the setting of setup a in the code reader. a=1-20
	PNAME=a,b	Name the SETUP a=1-20 b=(32 alphanumeric characters or less)
	WSETP	Save the SETUP name
	?plist	Output a list of SETUP names

7.17 General Operation

Term	Command	Description (*): Default
Initialization	SET=DFT	Reset to factory default
Save settings	WSETS	Save settings to internal flash memory
Digital I/O	CHATT=a	Debouncing time a=0: None a=1: 10[ms] (*) a=2: 20[ms]
	GOOUT=a	Set hold time of GO output (Hold off at next SYNC IN) a=100-20000 [ms] a=300 (*)
	NGOUT=a	Set hold time of NG output (Hold off at next SYNC IN) a=100-20000 [ms] a=300 (*)
Digital I/O signal assignment	DOUT1=a	Assign the signal on OUT1 port a=0: Disable a=1: GO (Read OK) (*) a=2: NG (Read NG) a=3: Ready a=4: Busy1 a=5: Busy2
	DOUT2=a	Assign the signal on OUT2 port a=0: Disable a=1: GO (Read OK) a=2: NG (Read NG) (*) a=3: Ready a=4: Busy1 a=5: Busy2
RESET	RESET	Reboot the reader.

7.18 Table of Character Code

		Upper Byte															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Lower Byte	0	NUL	DLE	SP	0	@	P	`	p								
	1	SOH	DC1	!	1	A	Q	a	q								
	2	STX	DC2	"	2	B	R	b	r								
	3	ETX	DC3	#	3	C	S	c	s								
	4	EOT	DC4	$	4	D	T	d	t								
	5	ENQ	NAK	%	5	E	U	e	u								
	6	ACK	SYN	&	6	F	V	f	v								
	7	BEL	ETB	'	7	G	W	g	w								
	8	BS	CAN	(	8	H	X	h	x								
	9	HT	EM	)	9	I	Y	i	y								
	A	LF	SUB	*	:	J	Z	j	z								
	B	VT	ESC	+	;	K	[	k	{								
	C	FF	FS	,	<	L	\	l	|							
	D	CR	GS	-	=	M	]	m	}								
	E	SO	RS	.	>	N	^	n	~								
	F	SI	US	/	?	O	_	o	DEL								

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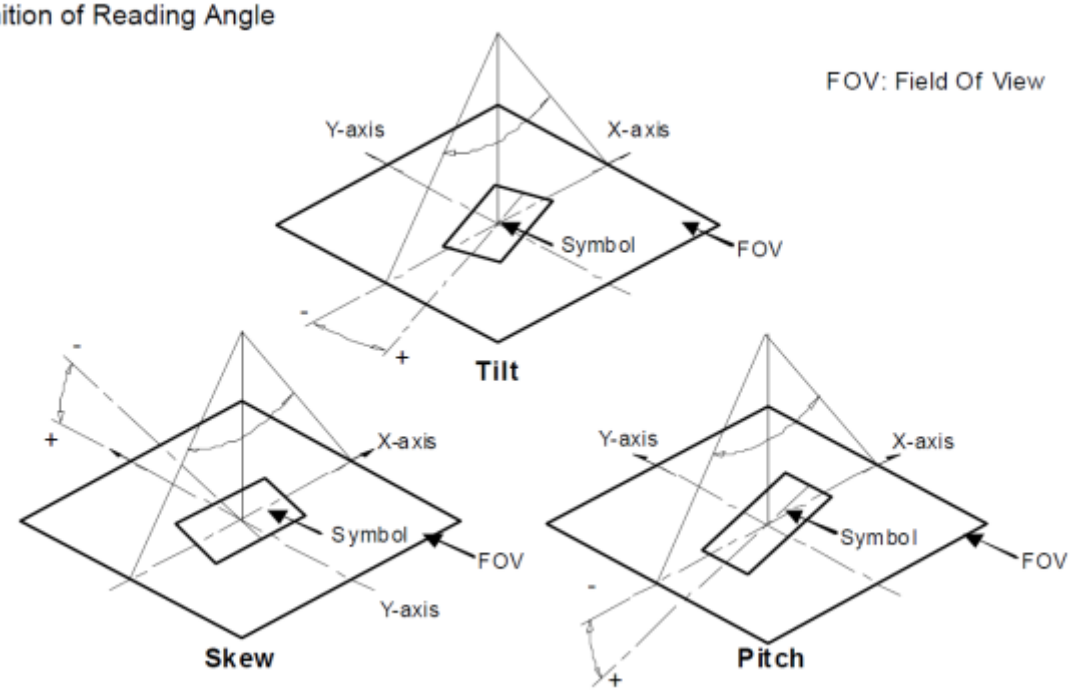
8 Specifications

8.1 Specifications

		MCR-F110R	MCR-F110U
Supported Symbols	Barcode	Code39, Code128 / GS1-128, Codabar(NW7) GS1 DataBar, ITF(Interleaved 2 of 5) JAN / UPC / EAN(JAN) , Code93	
	2D code	Data Matrix(ECC200), QR Code / Micro QR Code PDF417 / MicroPDF, AztecCode, CompositeCode	
Reading angle		PITCH: ±35° SKEW: ±35° TILT: 360°	
Minimum resolution		Barcode: 0.125 mm 2D code: 0.125 mm	
Image sensor	sensor	1/3-inch, Monochrome	
	Active pixels	752 (X) × 480 (Y)（Approx. 0.36 million pixels）	
Illumination		High-intensity red LED	
Pointer		Red LED	
Power rating	Supply voltage	DC 24 V ± 10%	DC 5 V ± 5%
	Current consumption	150 mA or less	500 mA or less
Digital input		Photo-coupler isolation x1 Input resistance: 2.2 kΩ ON voltage: 10.8 to 28 V OFF voltage: 0 to 0.8 V	-
Digital output		Photo-coupler isolation x2 Maximum rating: DC 30 V 50 mA	-
Interface		RS-232C(1200～115200 bps）	USB2.0(Full Speed)
Cable length		2m	
Environmental tolerance	Operating temperature	0 ～ 40 ℃	
	Operating humidity	35 ～ 85 %R.H. (No condensing)	
	Storage temperature	-20 ～ 60 ℃	
	Storage humidity	35 ～ 85 %R.H. (No condensing)	
	Vibration tolerance	10 to 55 Hz amplitude 1.5 mm (total width) X, Y, Z 1 hour for every direction (Note 1)	
	Enclosure rating	IP65	
	Ambient illuminance	Less than 10000 lx	
Dimensions		40(W)mm × 36(D)mm × 24(H)mm (without protrusions)	
Weight		Approx. 30 g (without the cable)	

Note 1: Not a guaranteed spec., but a test result.

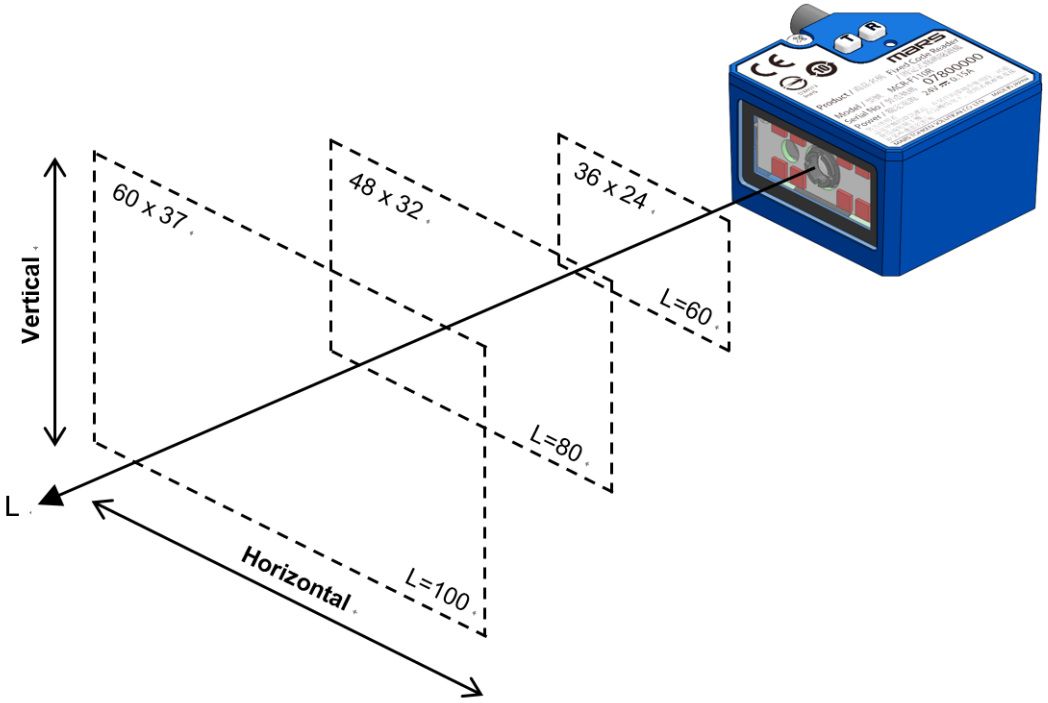
Definition of Reading Angle



8.2 Reading Specifications

8.2.1 Field of view

Reference: Center of Reading window (surface)



Unit: mm

Distance L	40	60	80	100
Horizontal	24	36	48	60
Vertical	17	24	32	37

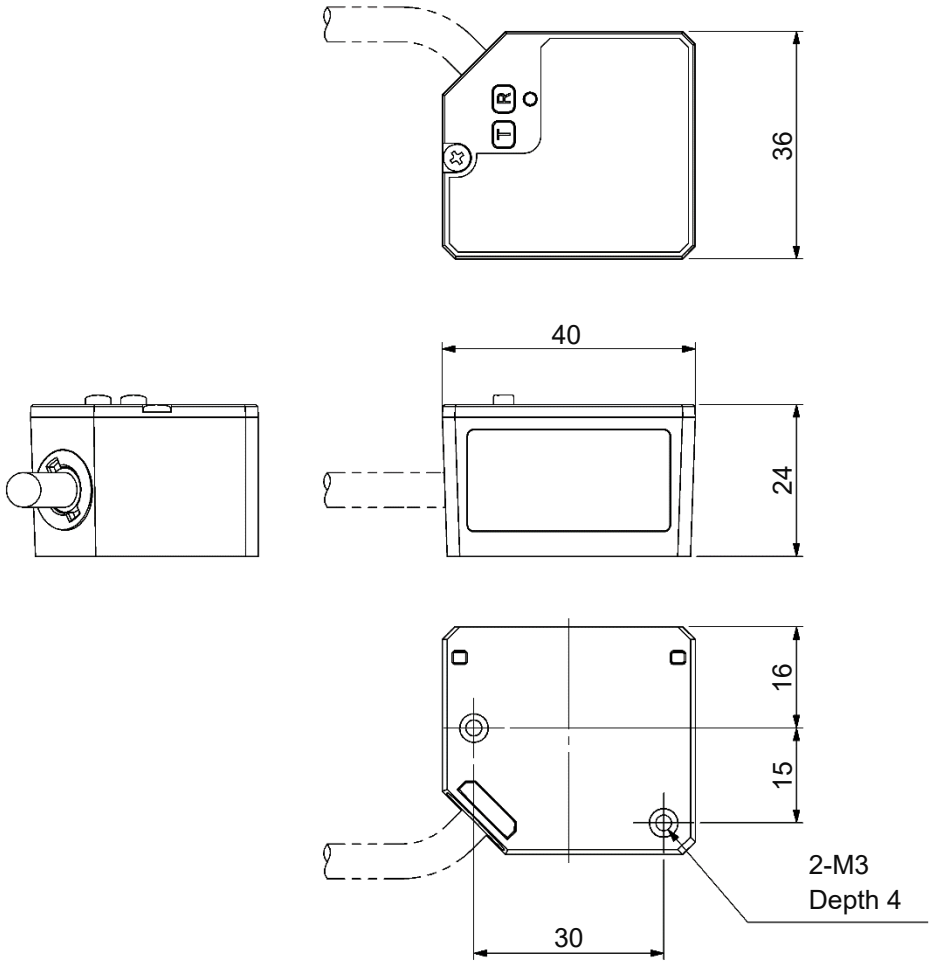
8.2.2 Reading range

Unit: mm

Narrow bar width / Cell size	Near	Far
Barcode 0.125	49	60
Barcode 0.167	42	90
Barcode 0.19	38	100
2D code 0.125	53	58
2D code 0.167	48	80
2D code 0.19	43	90

* Under the conditions of use of our specified symbols.

8.3 Dimensions

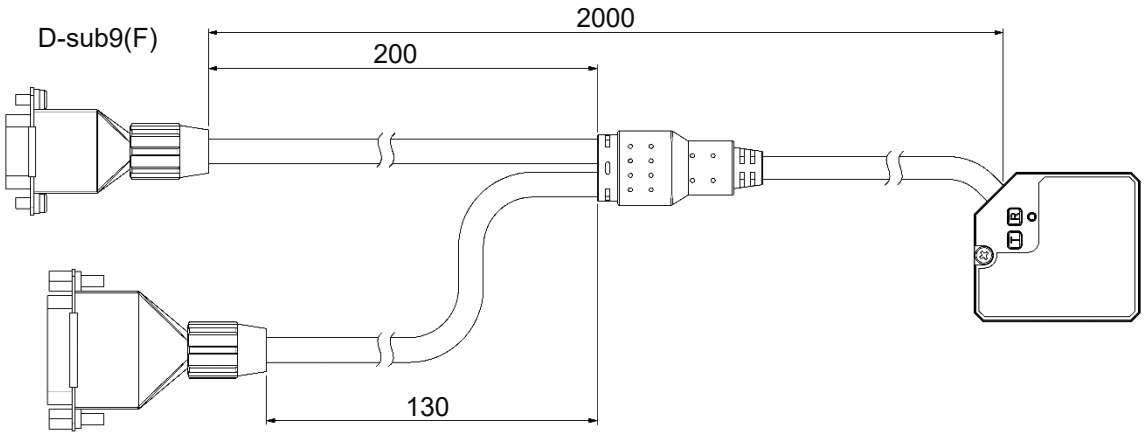


Unit: mm

WARNING

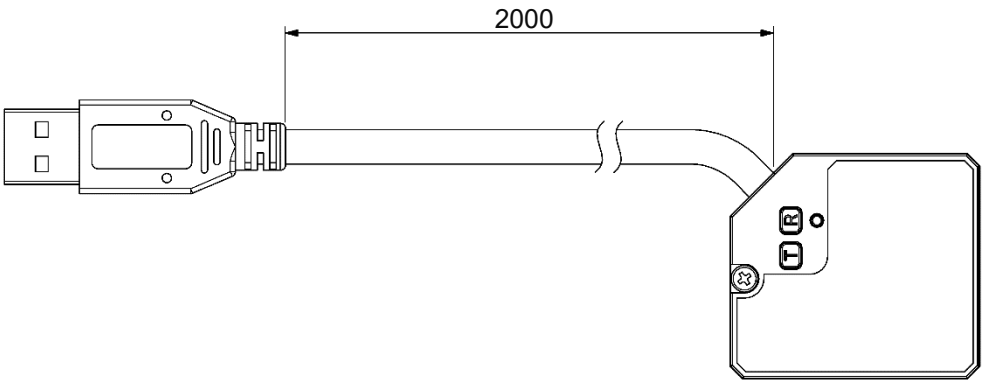
DO NOT tighten the screws more than the depth of mounting holes or the maximum tightening torque is as below.
M3: 0.7 [N·m]

RS-232C type



Cut the cable at the connector end when using cable with loose-wires.
Refer to 8.4 “Interface” for wire color.

USB type



Unit: mm

8.4 Interface

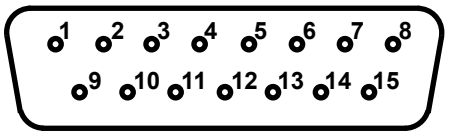
8.4.1 RS-232C type

(1) Power, I/O interface

Connector: D-sub 15pin (Male)

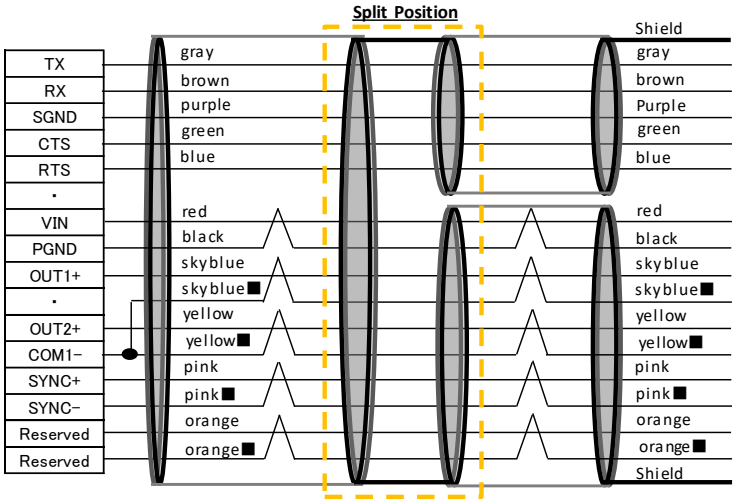
Lock screw: #4-40UNC (Female)

Connector pin assignment



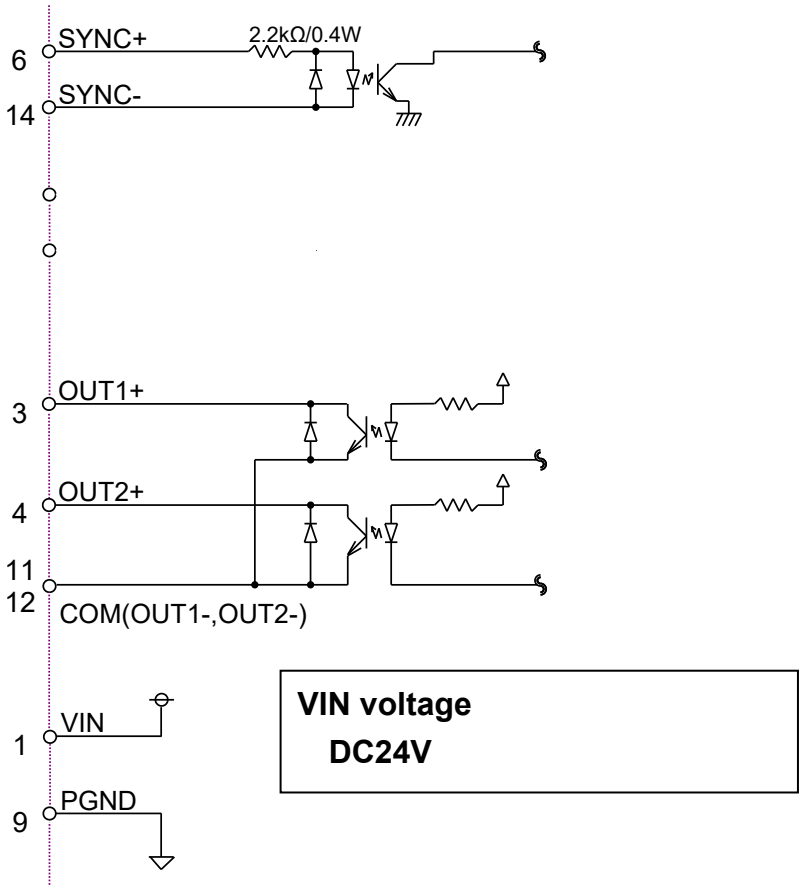
Pin	Color	Name	Function	Direction
1	Red	VIN	+24V	
2		N.C	Not connected	
3	Sky blue	OUT1+	Digital Output 1	Output
4	Yellow	OUT2+	Digital Output 2	Output
5		N.C	Not Connected	
6	Pink	SYNC+	SYNC Input (+ side of Photo coupler)	Input
7	Orange	Reserved	Reserved (Do Not Connect)	
8		N.C	Not Connected	
9	Black	PGND	Power Ground	
10		N.C	Not Connected	
11	Sky blue ■	OUT1-	Digital Output Common (Connected internally)	Output
12	Yellow ■	OUT2-		
13		N.C	Not Connected	
14	Pink ■	SYNC-	SYNC Input (- side of Photo coupler)	Input
15	Orange ■	Reserved	Reserved (Do Not Connect)	
Case	Shield		Connect the shield to the ground as FG.	

Connection table for loose-wires



*Ground the shield as FG.

Internal equivalent circuit



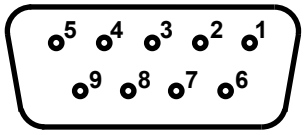
VIN voltage
DC24V

(2) RS-232C interface

Connector: D-sub 9pin (Female)

Lock screw: #4-40UNC (Male)

Connector pin assignment



Pin	Color	Name	Function	Direction
1		N.C	Not Connected	
2	Gray	TXD	RS-232C Transmit Data	Output
3	Brown	RXD	RS-232C Receive Data	Input
4		N.C	Not Connected	
5	Purple	SGND	Signal Ground	
6		N.C	Not Connected	
7	Green	CTS	RS-232C Clear To Send	Input
8	Blue	RTS	RS-232C Request To Send	Output
9		N.C	Not Connected	

8.4.2 **USB type**

USB connector Type-A

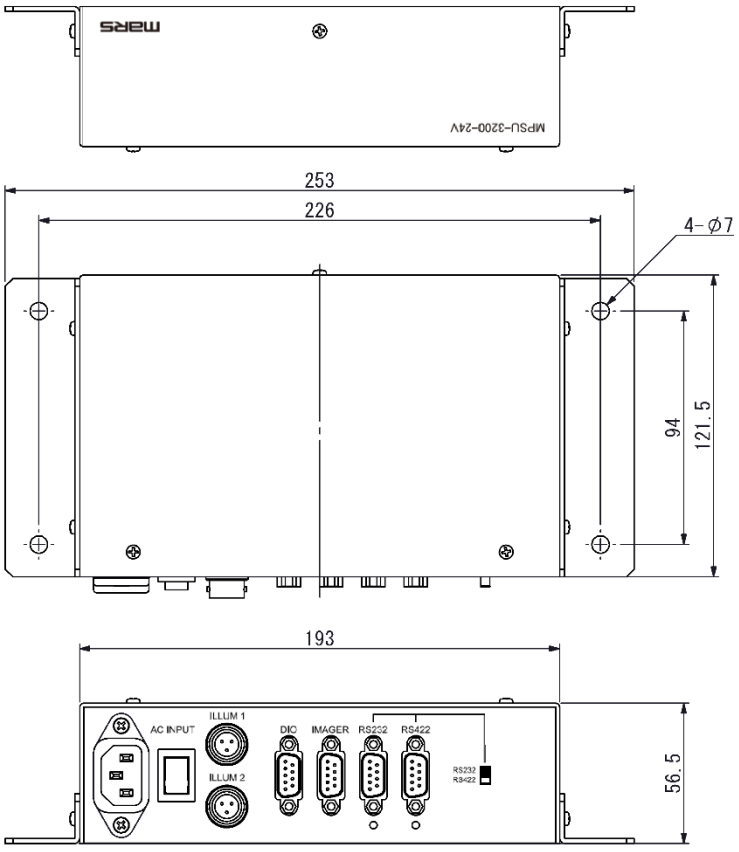
9 Peripherals

9.1 Peripheral for MCR-F110R

9.1.1 Power supply unit MPSU-3200-24V

Item	Value
Input power	AC100V to 240V
Output voltage	24Vdc（IMAGER 24Vdc）
Output power	64W

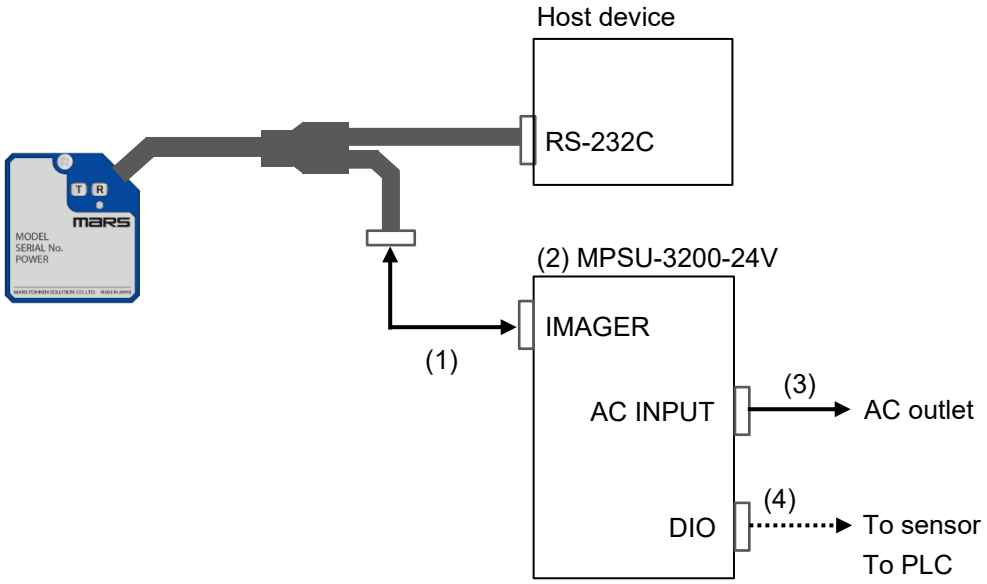
Dimensions (Unit: mm)



9.1.2 **Example of device configuration**

The followings are the conditions that are used for this configuration:

- Use an external sensor to input SYNC
- Use digital outputs of MCR-F110R



No.	Product name	Model name	Note
(1)	MPSU-3200 relay cable	AA000343	Optional
(2)	Power supply unit	MPSU-3200-24V	Optional
(3)	AC power cable		Bundled to MPSU-3200-24V
(4)	DIO connector (Dsub9P Male, chassis)		Bundled to MPSU-3200-24V

10 Troubleshooting

If any problem occurs during the operation, try the following methods to recover.
The host device which is the access point may cause a problem. Refer to the instruction manual of your host device.

10.1 MCR-F110 does not start up or cannot communicate with a PC.

- **Is power supply voltage within the specifications?**

  Make sure that the power supply voltage of MCR-F110 (RS-232C type) is DC24V +/- 10%.
  If the voltage is out of the specification range, the reader may not operate properly or may be damaged.

- **Is MCR-F110 properly connected to the host device?**

  Make sure that the reader is properly connected to the host device.
  In the RS-232C interface, all communication conditions including the baud rate must be the same.

- **Is the format of serial command correct?**

  Serial commands are always accepted when the reader is in standby state.

  Add a carriage return code [CR] at the end of the serial command.

  Push the Enter key to enter serial command in the terminal software.

  After configuring the reader, send the command WSETS to save the settings to internal flash memory. If you turn off the reader without saving, the settings will not be saved.

  Take a command transmission interval of 500 ms or more to avoid unwanted errors during the reader setting.

10.2 Symbol cannot be decoded

- **Is the symbol setting correct?**

  Make sure that the symbology and number of digits settings are correct.
  If the symbol type or number of digits is out of the setting, the reader can not read.

- **Are the reading distance and the print quality suitable?**

  Set the symbol within the reading specification range.
  If the symbol is damaged or the print quality is low, the reader may not decode even within the reading range.

- **Is the reading window clean?**

  The quality of captured images is affected by the condition of the reading window. Dust, dirt, etc. on the reading window may alter the reading performance. Clean with a cleaner or similar non-abrasive method.

- **Is the print quality of the symbol good?**

  Make sure the quality of the symbol (1D: Module width and Wide/Narrow bar width ratio, 2D: Cell size) meet the standards.

10.3 Others

Press the Reset key to perform a hardware reset. After the reader is turned off, it will restart.

The cable of MCR-F110R is shielded cable. Ground the shield when using the cable with loose-wires.

If there is anything unclear or if you have any problems that can not be resolved by the above check, contact our sales representative.

<<Memorandum>>

Warranty Obligations

Exclusive of the cases with special mentions in quotations, contracts or specifications, MARS TOHKEN SOLUTION warrants this product as follows.

1. Term of Warranty
   One year from the date of purchase.

2. Warranty Coverage
   MARS TOHKEN SOLUTION shall repair or replace the product at no charge if any defects in the Products are suspected to be due to design or manufacturing error.
   This warranty does not cover, and MARS TOHKEN SOLUTION will not be held liable, for any damages incurred by the use of this product.
   The above warranty shall not apply to any Product;
   (1) which has not been used and maintained in accordance with the operating instructions,
   (2) which has been repaired or altered by anyone other than MARS TOHKEN SOLUTION authorized personnel,
   (3) which has been subject to unusual physical or electrical stress, misuse, abuse, negligence, natural disasters, fire, pollution etc.,
   (4) other defects that may arise through no fault of MARS TOHKEN SOLUTION.

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